

Large-scale Sequence Search using Exact Indices (k-mer sets as de Bruijn graphs)

A fundamentally different approach

Our initial idea — the Bloom Filter is limiting.
What can we get by replacing it with a *better* AMQ

A General-Purpose Counting Filter: Making Every Bit Count

Prashant Pandey, Michael A. Bender, Rob Johnson, and Rob Patro
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SIGMOD 2017

Interesting observation
about patterns of k-mer occurrence

Rainbowfish: A Succinct Colored de Bruijn Graph Representation*

Fatemeh Almodaresi¹, Prashant Pandey², and Rob Patro³

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WABI 2017

Mantis: A Fast, Small, and Exact Large-Scale Sequence-Search Index

Prashant Pandey¹, Fatemeh Almodaresi¹, Michael A. Bender¹, Michael Ferdman¹, Rob Johnson^{2,1}, and Rob Patro¹

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robj@vmware.com

“I bet we can exploit
that for large-scale search”

The CQF

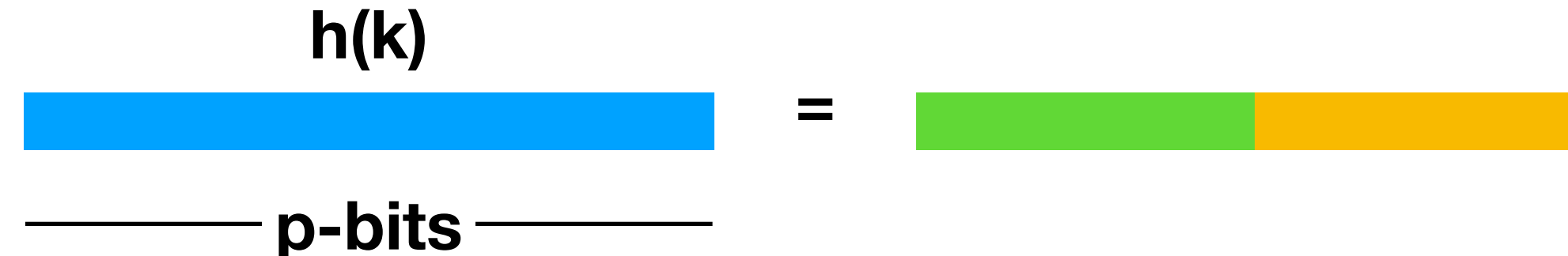
Approximate *Multiset* Representation

	0	1	2	3	4	5	6	7
occupieds	0	1	0	1	0	0	0	1
runends	0	0	0	1	0	1	0	1
remainders		$h_1(a)$	$h_1(b)$	$h_1(c)$	$h_1(d)$	$h_1(e)$		$h_1(f)$

← 2^q →

Works based on quotienting* & fingerprinting keys

Let k be a key and $h(k)$ a p -bit hash value



Clever encoding allows low-overhead storage of element counts
(use *key* slots to store *values* in base 2^r-1 ; smaller values $\Rightarrow$ fewer bits)

Careful engineering & use of efficient rank & select to resolve collisions leads to a **fast, cache-friendly** data structure

* Idea goes back at least to Knuth (TACOP vol 3)

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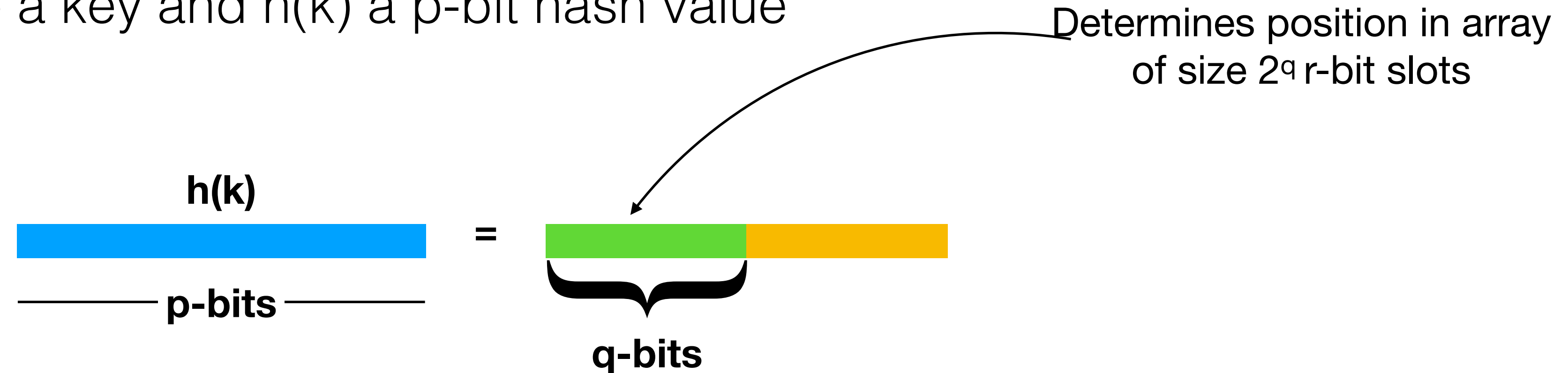
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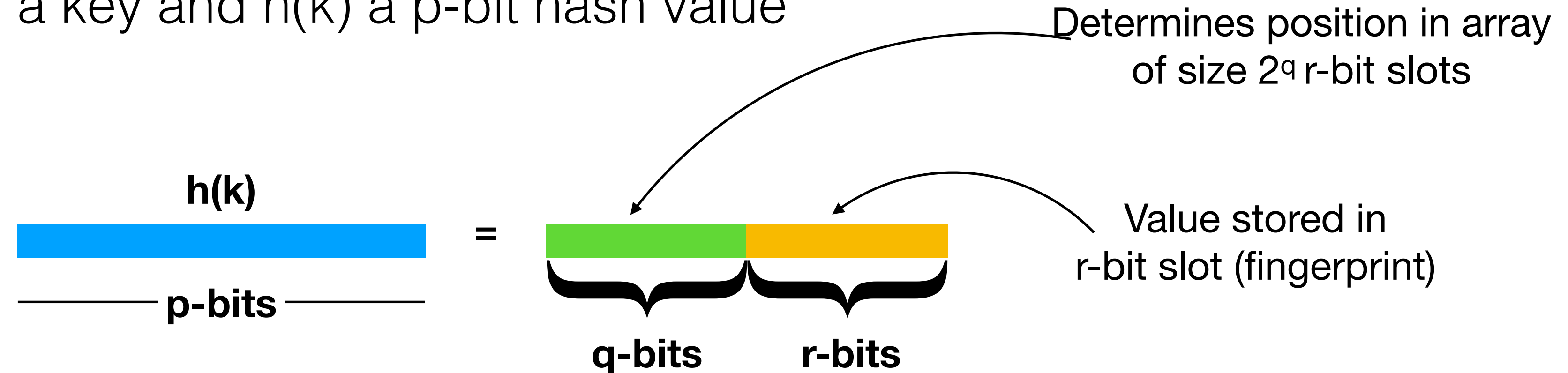
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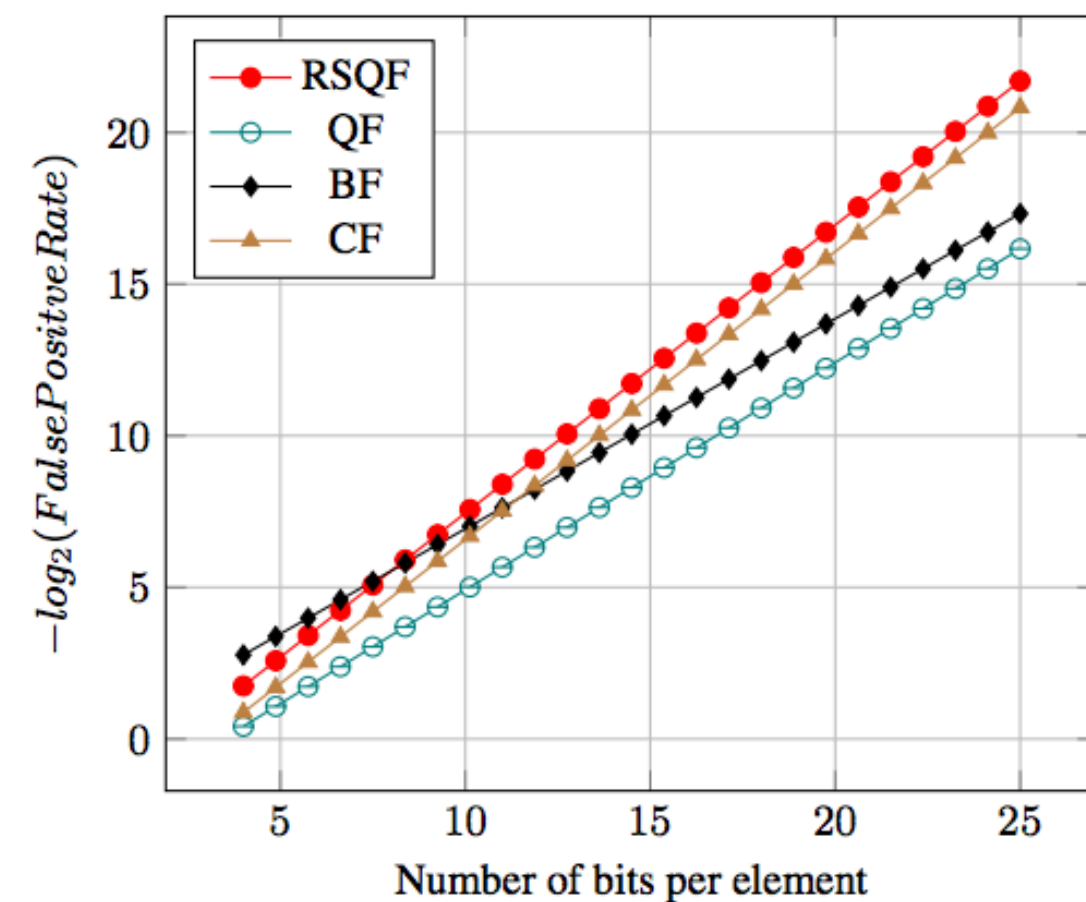
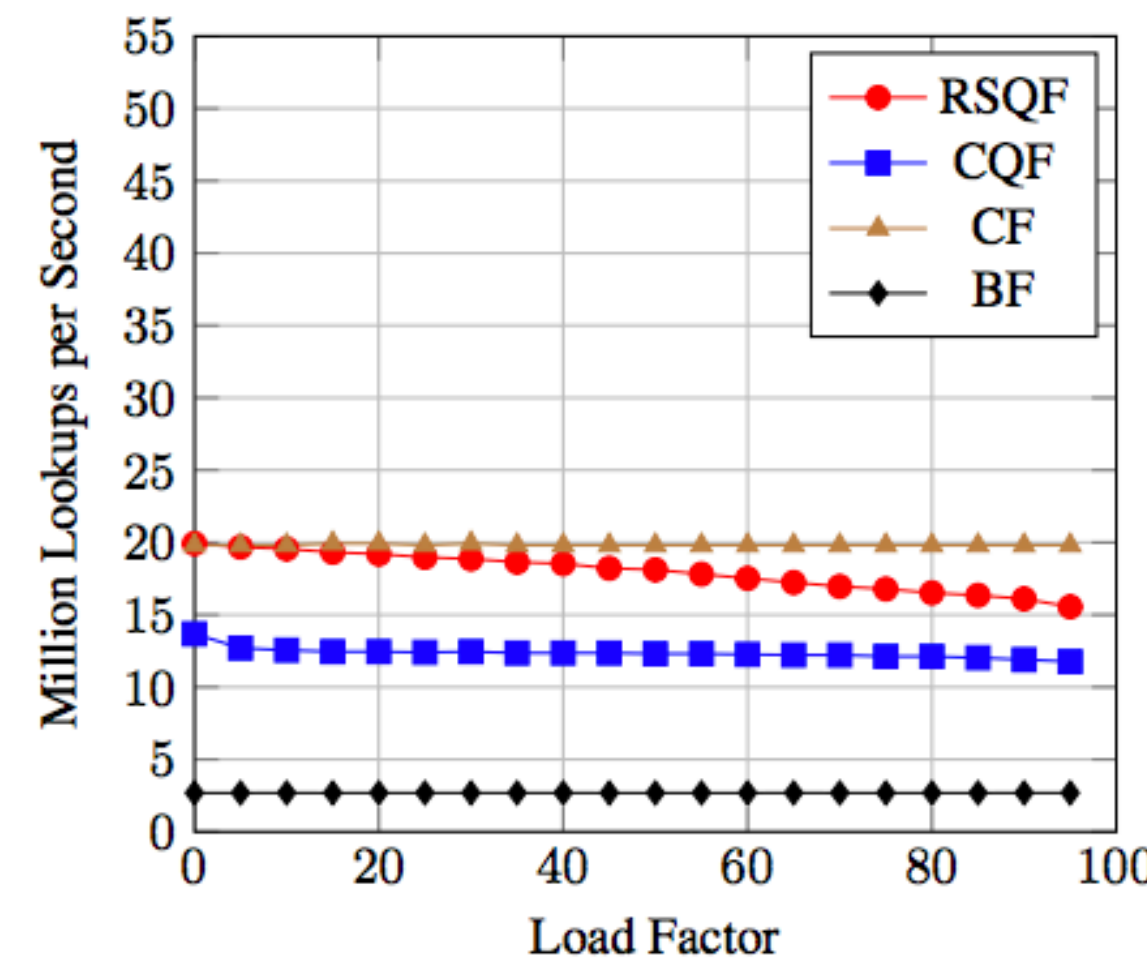
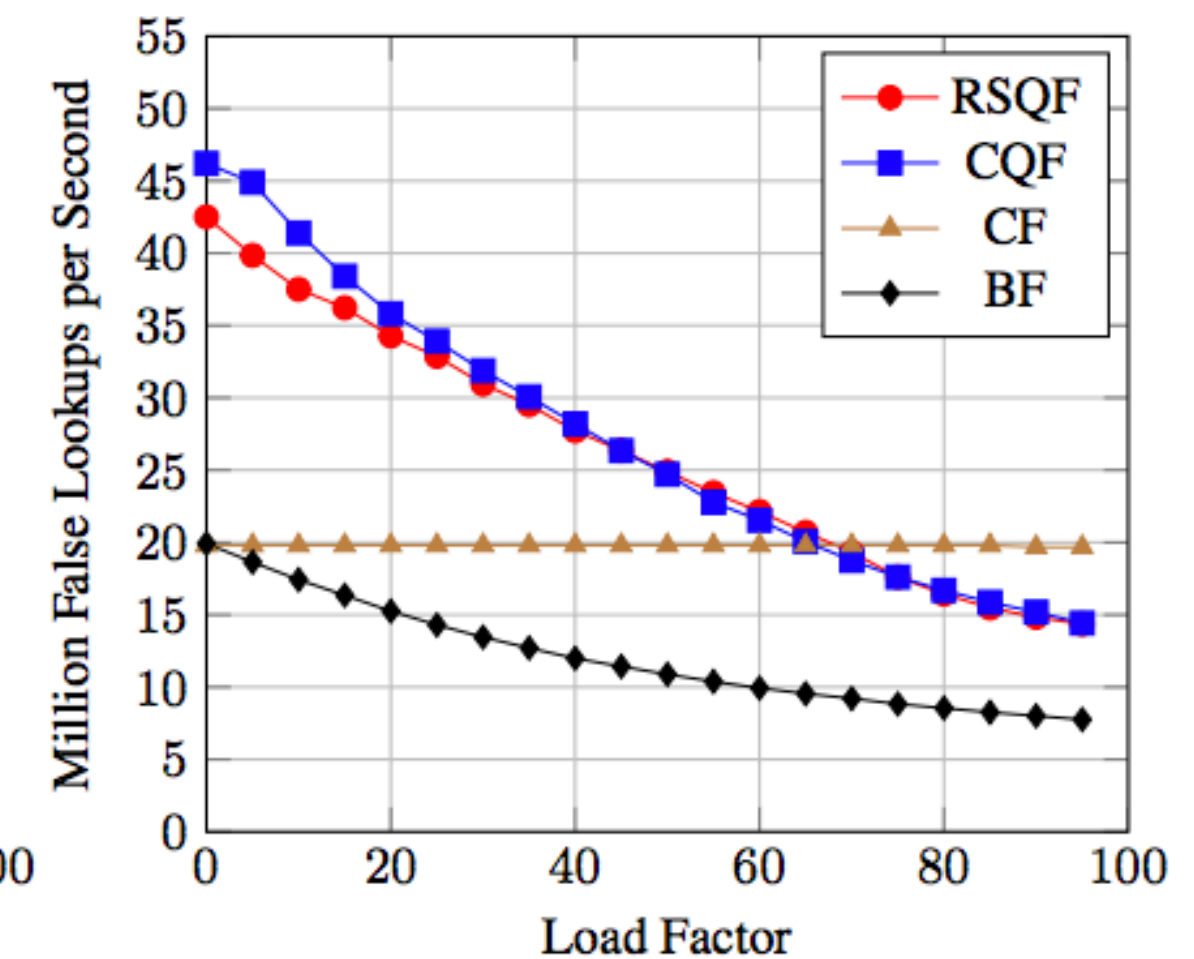


Figure 4: Number of bits per element for the RSQF, QF, BF, and CF. The RSQF requires less space than the CF and less space than the BF for any false-positive rate less than $1/64$. (Higher is better)



(b) Successful lookups.



(c) Uniformly random lookups.

Mantis

Observation 1 : If I want to index N k-mers over E experiments, there are $\leq \min(N, 2^{|E|})$ possible distinct “patterns of occurrence” of the k-mers, there are usually *many* fewer.

Observation 2 : These patterns of occurrence are *far* from uniform. Specifically, k-mers don't occur independently, occurrences are *highly correlated*.

Why?

Mantis

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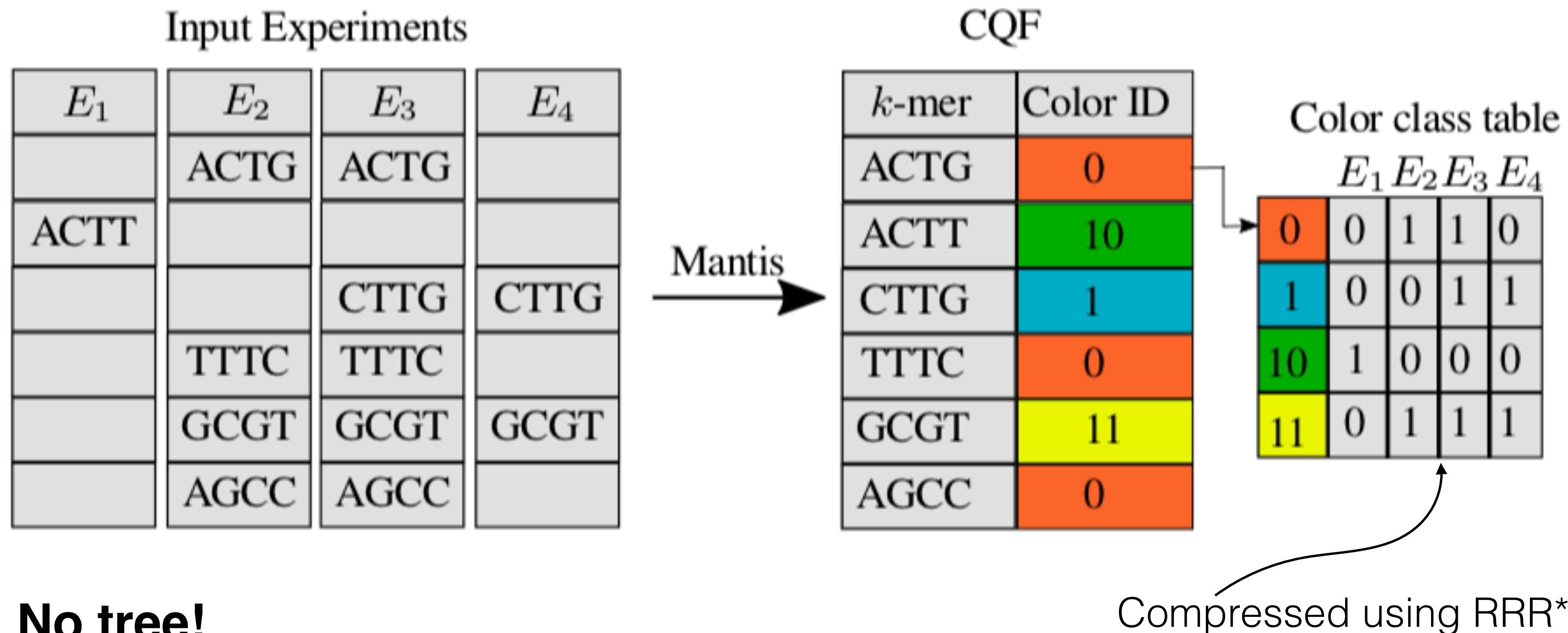
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What if we add a layer of indirection: Store each distinct pattern (color class) only once. *label* each pattern with an index, s.t. frequent patterns get small numbers (think Huffman encoding)

David Wheeler approves ... we think.

<https://github.com/splatlab/mantis>

The Mantis Index: Core Idea



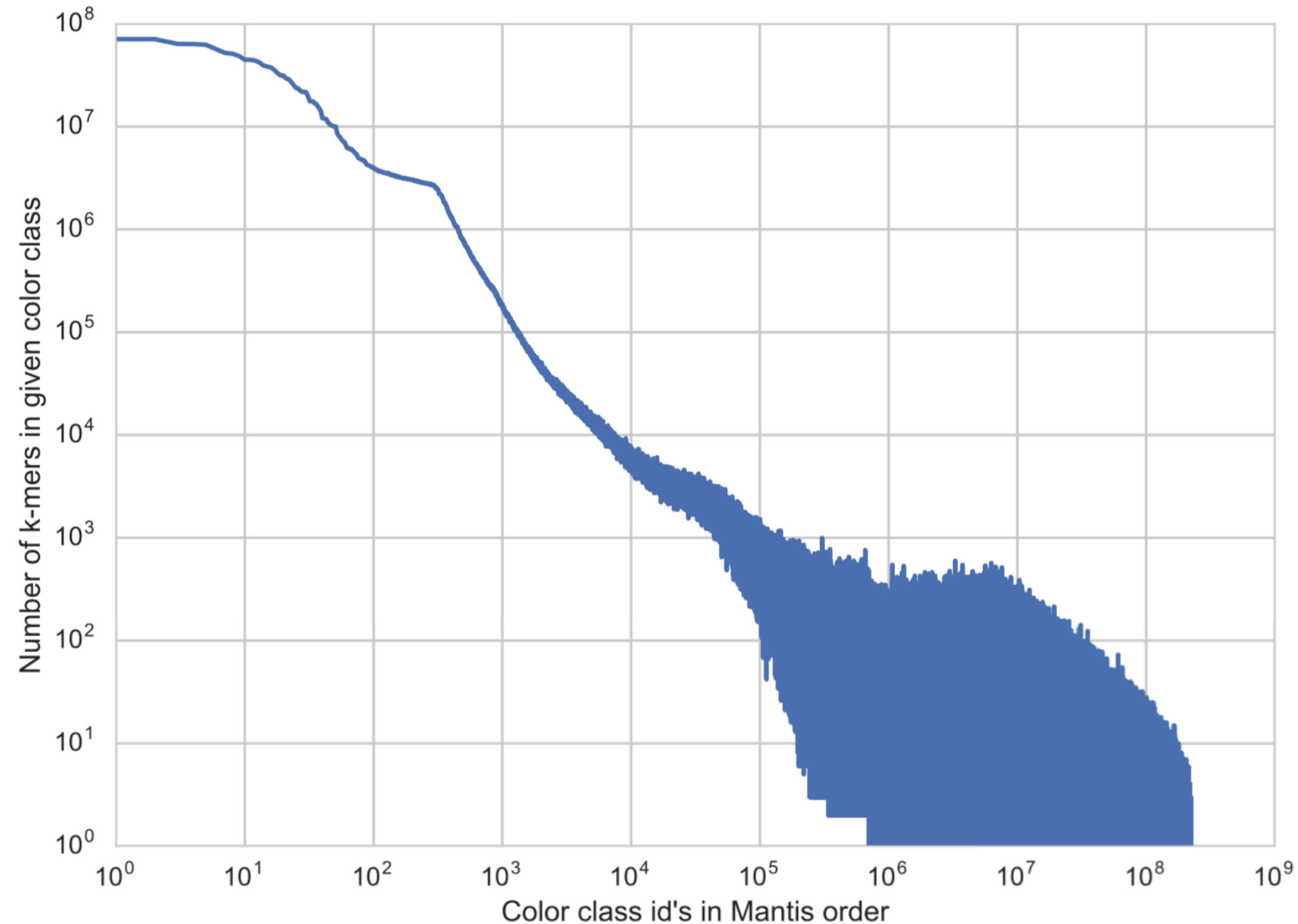
No tree!

- Build a CQF for each input experiment (can be different sizes, since CQFs of different sizes are mergeable)
- Combine them via multi-way merge
- CQF : **key** = k -mer, **value** = color class ID
- *Estimate* a good ordering of color class IDs from first few million k -mers

*Raman, et al. (2002). Succinct indexable dictionaries with applications to encoding k -ary trees and multisets. In Proceedings of the thirteenth annual ACM-SIAM symposium on Discrete algorithms, pages 233–242.

Why does this work?

The distribution of k-mers / color class is *highly skewed*



~3.7 Billion k-mers from ~2,600 distinct sequencing experiments

Assign smaller labels to frequent colors & use fewer bits.

Label	Equivalence Class	Freq.
0	0110010101	3
1	0011110000	1
2	1111110101	2
3	1010101011	1
4	1110000001	1

Color Matrix
0
1
2
0
2
3
4
0

Equivalence Bitvector

000	001	010	000	010	011	100	000
-----	-----	-----	-----	-----	-----	-----	-----

Label Bitvector

Same as idea from Rainbowfish Almodaresi et al. 2016
This idea is briefly discussed in BFT paper

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0 10 1 0 1 11 100 0

Label Bitvector

000 010 001 000 001 011 100 000

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Mantis : Comparing to SSBT

Construction Time — How long does it take to build the index?

Index Size — How large is the index, in terms of storage space?

Query Performance — How long does it take to execute queries?

Result Accuracy — How many FP positives are included in query results?

Bonus: If the remainder + quotient bits = original key size & we use an invertible hash, the CQF is *exact*.

Mantis is compact enough that we can *exactly* rather than *approximately* index the k-mers in our experiment set.

This lets us ask useful questions about how other approaches perform.

Mantis : Construction Time & Index Size

Indexed 2,652 human RNA-seq (gene expression) experiments
~4.5TB (GZip compressed) of data

Table 1. Time and Space Measurement for Mantis and SSBT

	Mantis	SSBT
Build time	16 hr 35 min	97 hr
Representation size	32 GB	39.7 GB

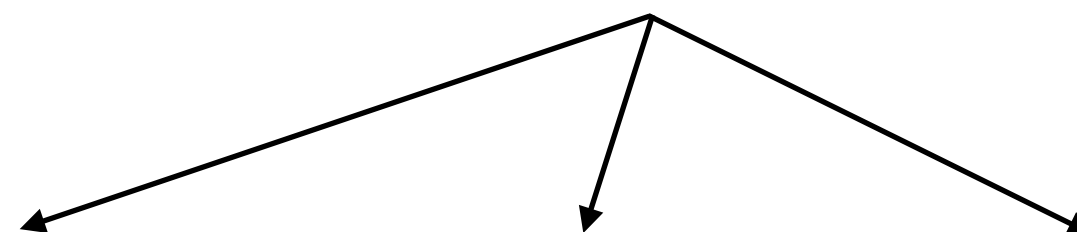
- Mantis can be constructed ~6x faster than a comparable SSBT
- The final Mantis representation is ~20% smaller than the comparable SSBT representation.

Note: both results assume you already have per-experiment AMQs (either Bloom Filters or CQFs)

Mantis : Query Speed

Querying for the presence of randomly selected genes across all 2,652 experiments.

θ threshold for SSBT query



	Mantis	SSBT (0.7)	SSBT (0.8)	SSBT (0.9)
10 Transcripts	25 s	3 min 8 s	2 min 25 s	2 min 7 s
100 Transcripts	28 s	14 min 55 s	10 min 56 s	7 min 57 s
1000 Transcripts	1 min 3 s	2 hr 22 min	1 hr 54 min	1 hr 20 min

- Mantis is ~6 — 109x faster than (in memory) SSBT

Note: Mantis doesn't require a θ threshold for queries, though one can be applied *post hoc*.

A Mantis query returns, for each experiment containing at least one query k-mer, the *fraction* (true θ) of query k-mers contained in the experiment.

Mantis : Query Quality

Querying for the presence of randomly selected genes across all 2,652 experiments. SSBT $\theta = 0.8$

	Both	Only Mantis	Only SSBT	Precision
10 Transcripts	2,018	19	1,476	0.577
100 Transcripts	22,466	146	10,588	0.679
1000 Transcripts	160,188	1,409	95,606	0.626

“Both” means the number of those experiments that are reported by both Mantis and SSBT. “Only Mantis” and “Only SSBT” mean the number of experiments reported by only Mantis and only SSBT. All three query benchmarks are taken from [Table 2](#) for $\theta = 0.8$.

- Recall : Mantis is exact! Returns *only* experiments having $\geq \theta$ fraction of the query k-mers.

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Due to a small number of corrupted SSBT filters — able to discover this b/c of Mantis' exa

Some Remaining Challenges

- It improves greatly upon existing solutions; takes a different approach
- We demonstrate indexing on the order of 10^3 experiments, we really want to index on the order of $10^5 - 10^6$
- Can be made approximate while providing strong bounds :

Theorem 1. *A query for q k -mers with threshold θ returns only experiments containing at least $\theta q - O(\delta q + \log n)$ queried k -mers w.h.p.*

but maybe not enough

Key Observation:

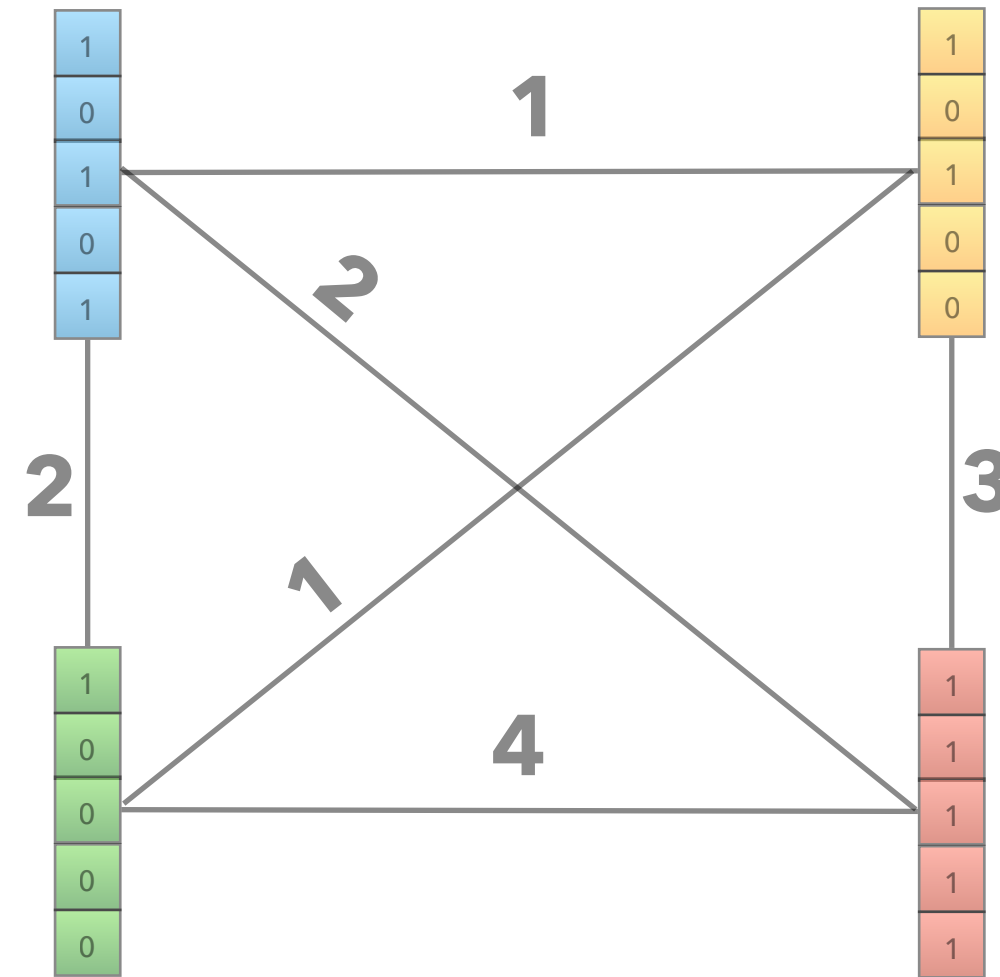
- K -mers grow at worst linearly
- Color classes increase super-linearly

Need a **fundamentally better** color class encoding; exploit *coherence* between rows of the color class matrix

Consider the following color class graph

Each color class is a vertex

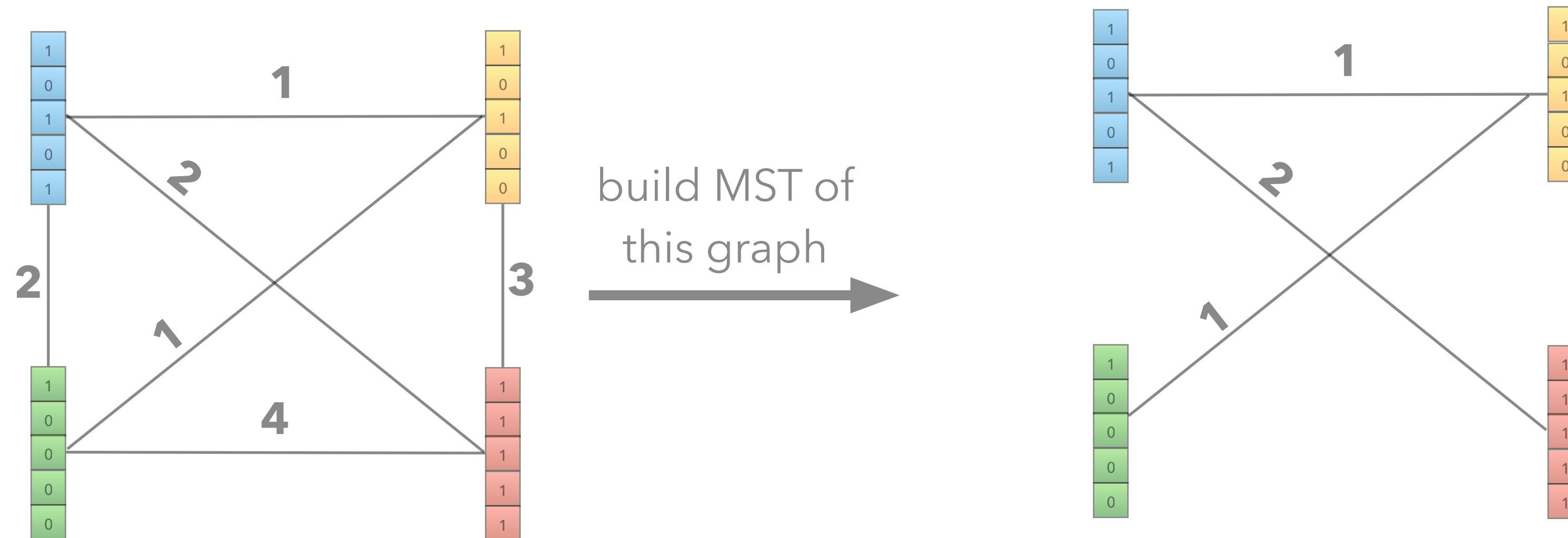
Every pair of color classes is connected by an edge whose weight is the **hamming distance** between the color class vectors



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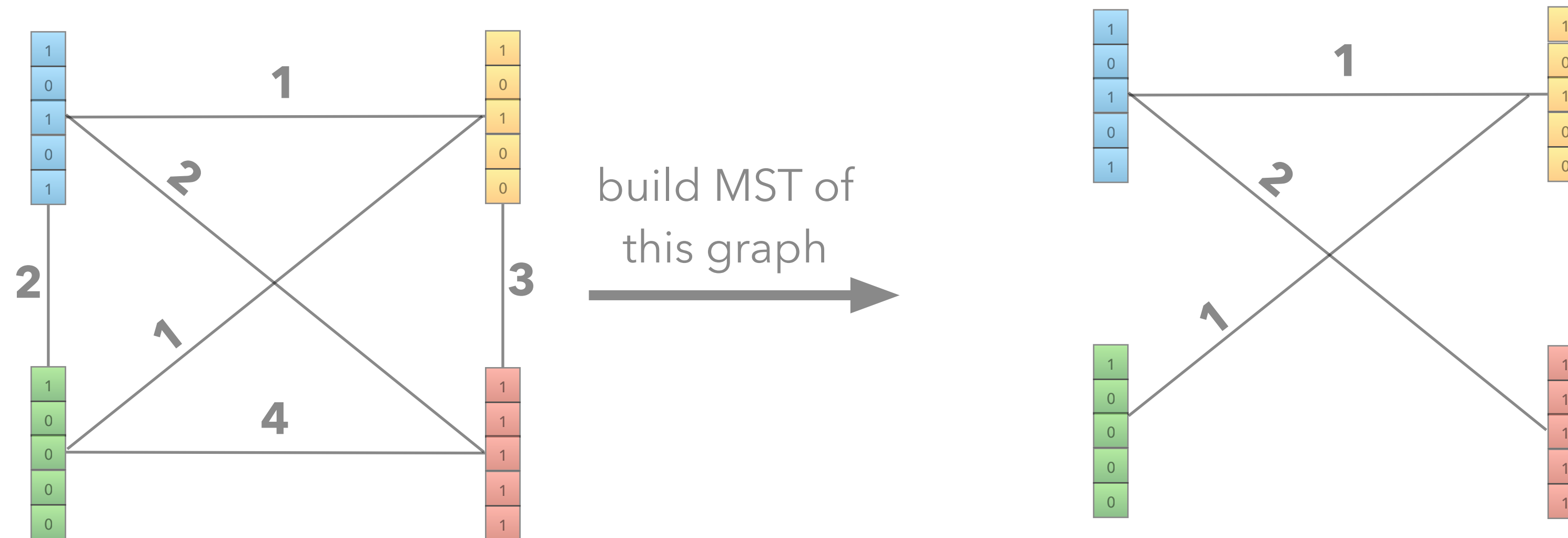
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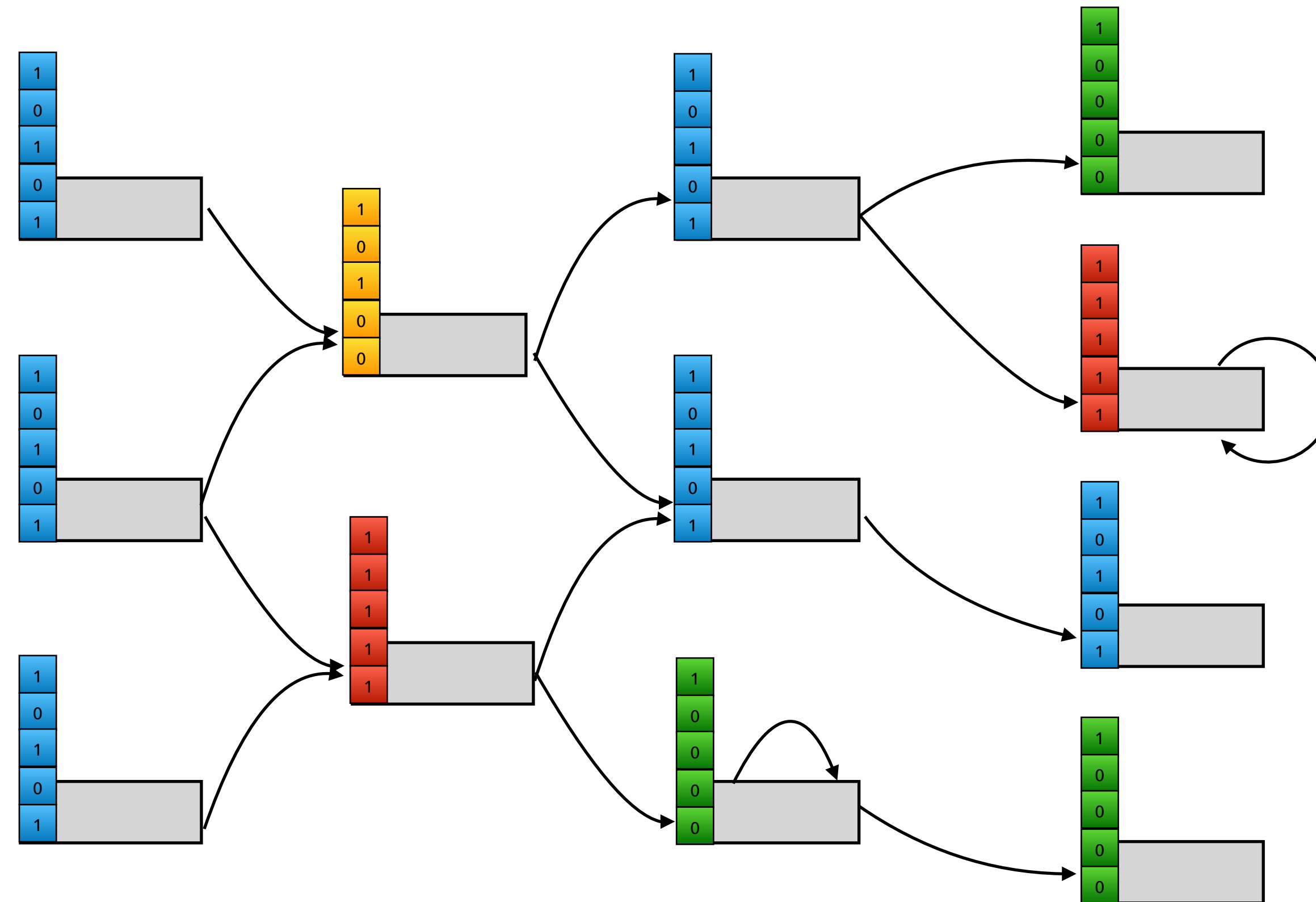
Unfortunately:

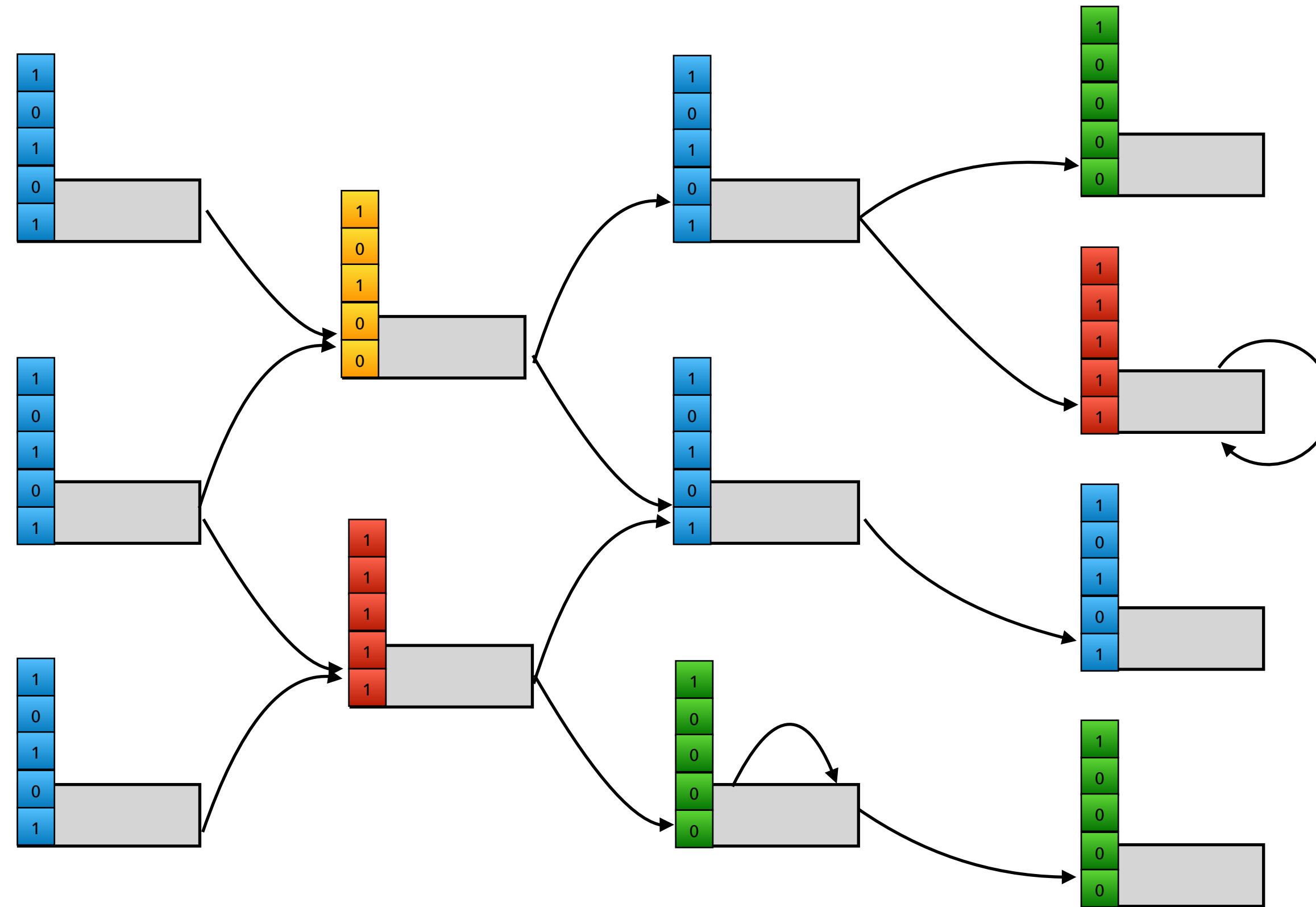
- 1) There are *many* color classes (full graph too big)
- 2) They are high-dimensional (# of experiments), neighbor search is very hard (LSH scheme seem to work poorly)

Mantis implicitly represents a colored dBG

Each CQF key represents a kmer → can explicitly query neighbors

Each k-mer associated with color class id → vector of occurrences





Use the **de Bruin graph** (dBG) as an efficient guide for near-neighbor search in the space of color classes!

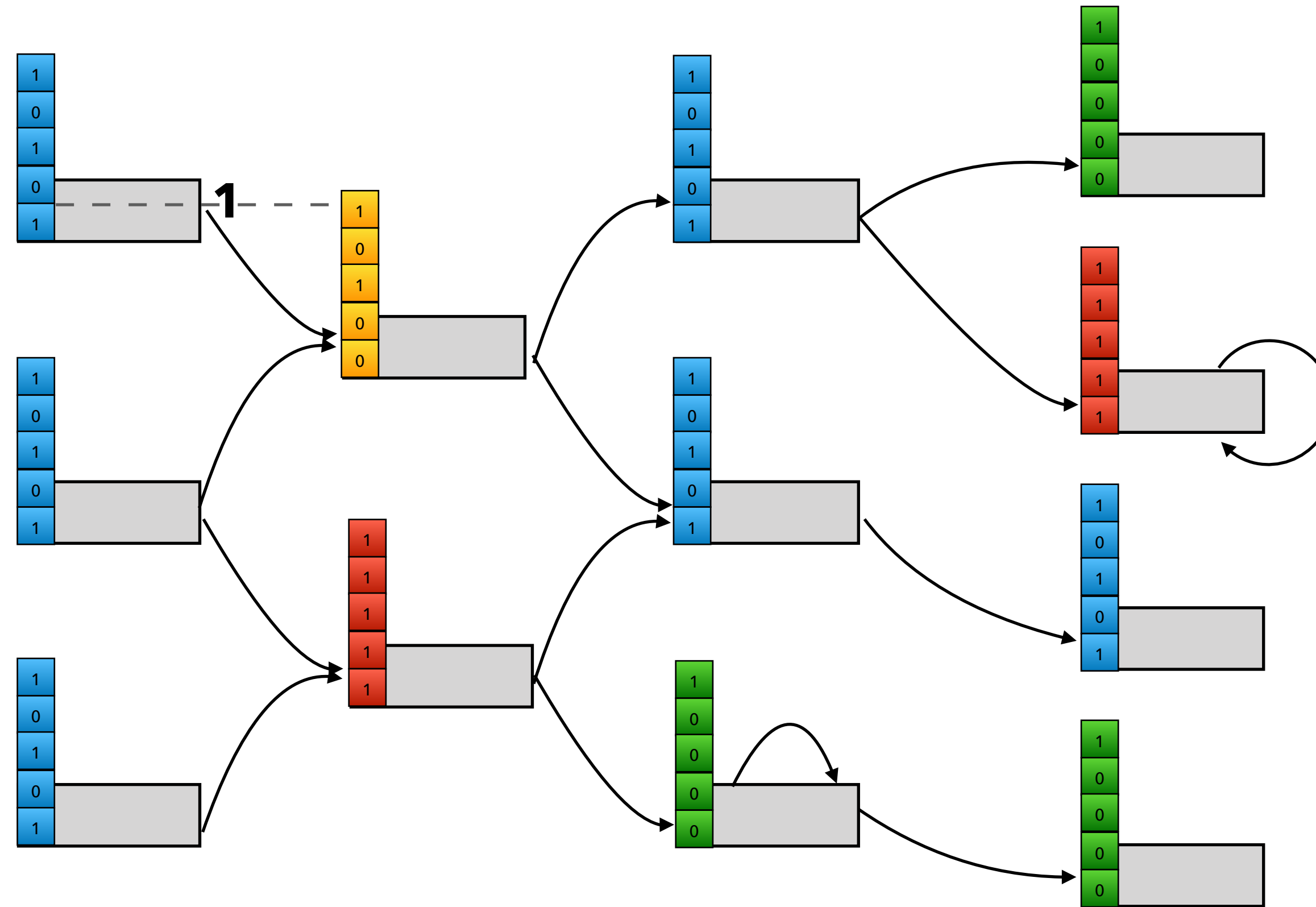
dBG common in genomics. Nodes u, v are k -mers & are *adjacent* if $k-1$ suffix of u is the same as $k-1$ prefix of v

1
0
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0

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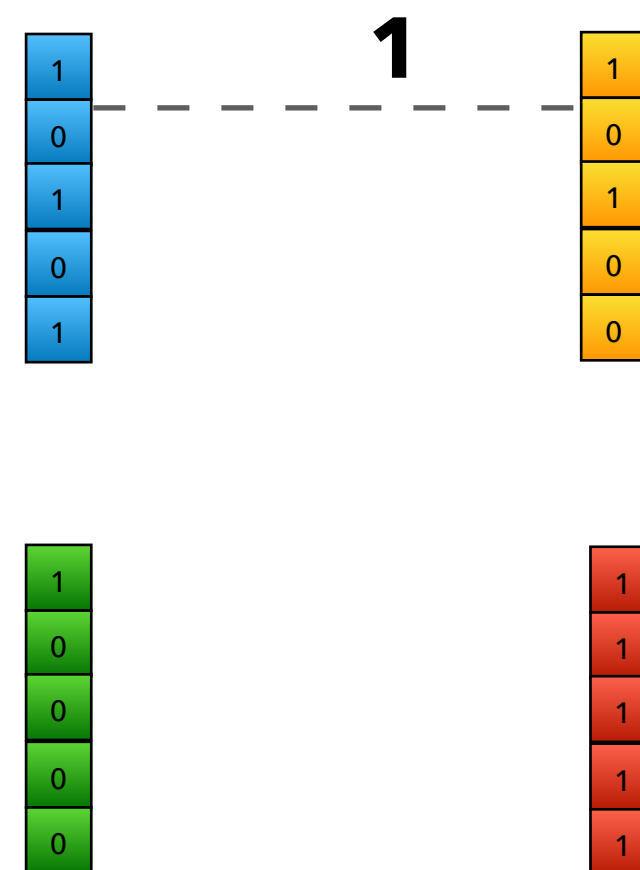
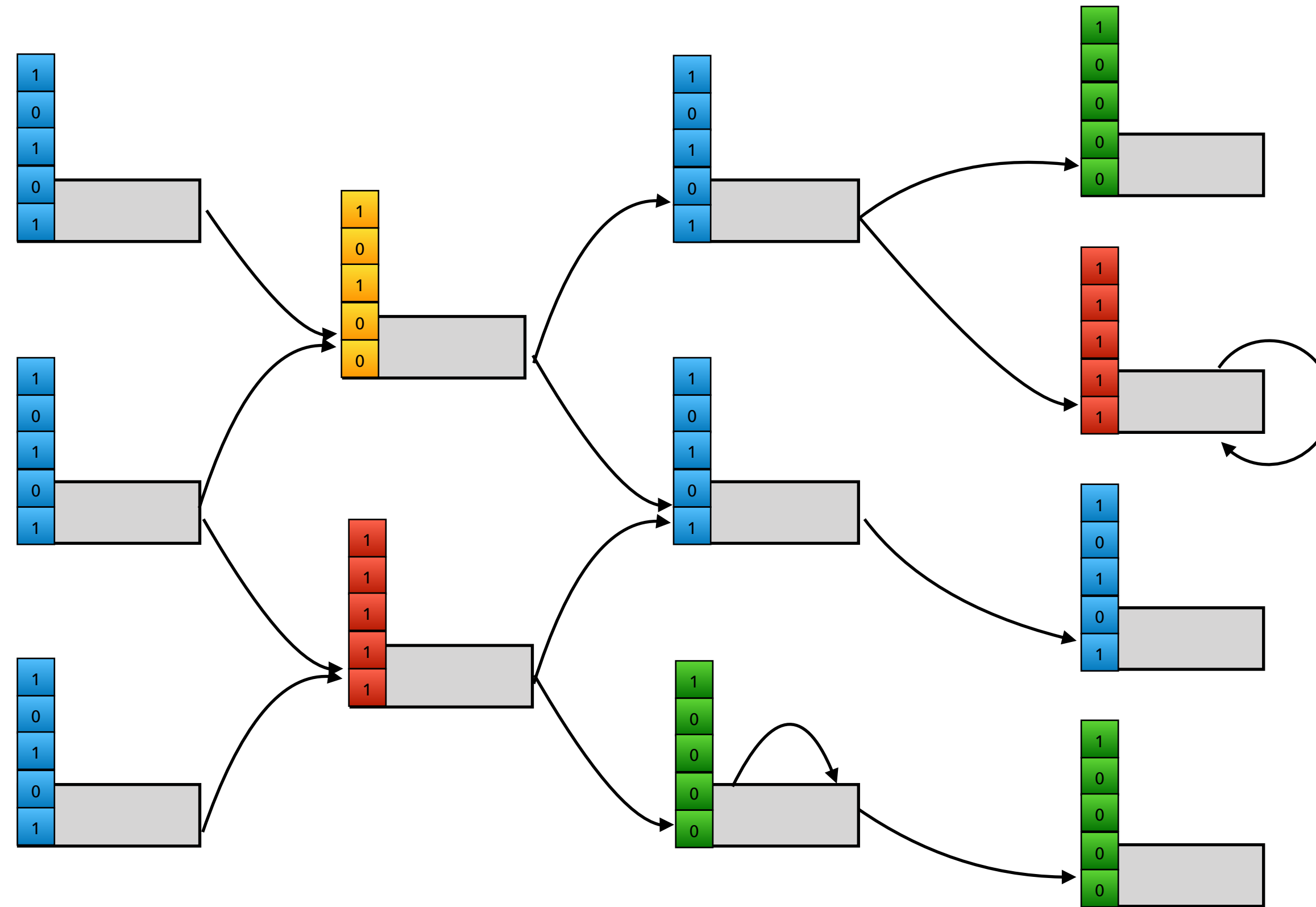
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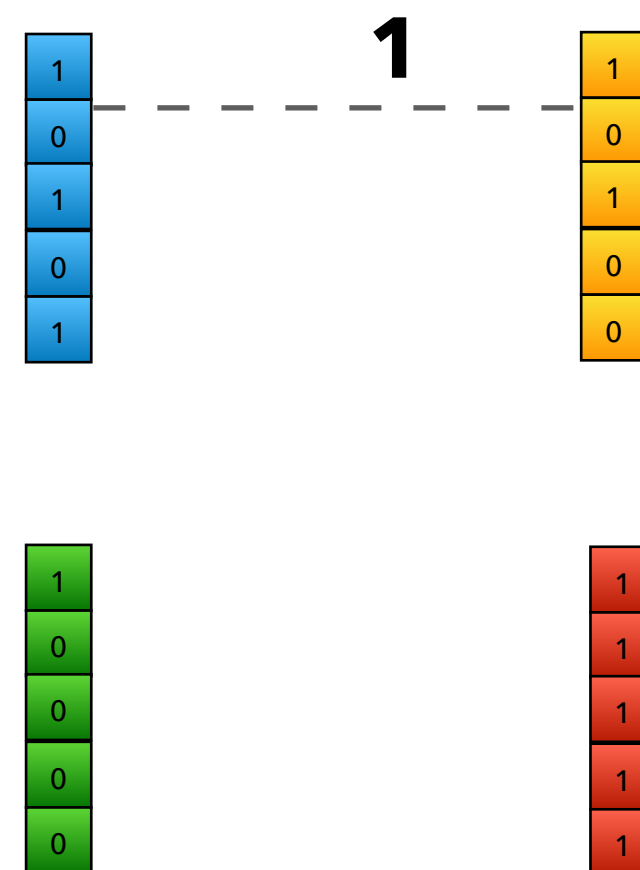
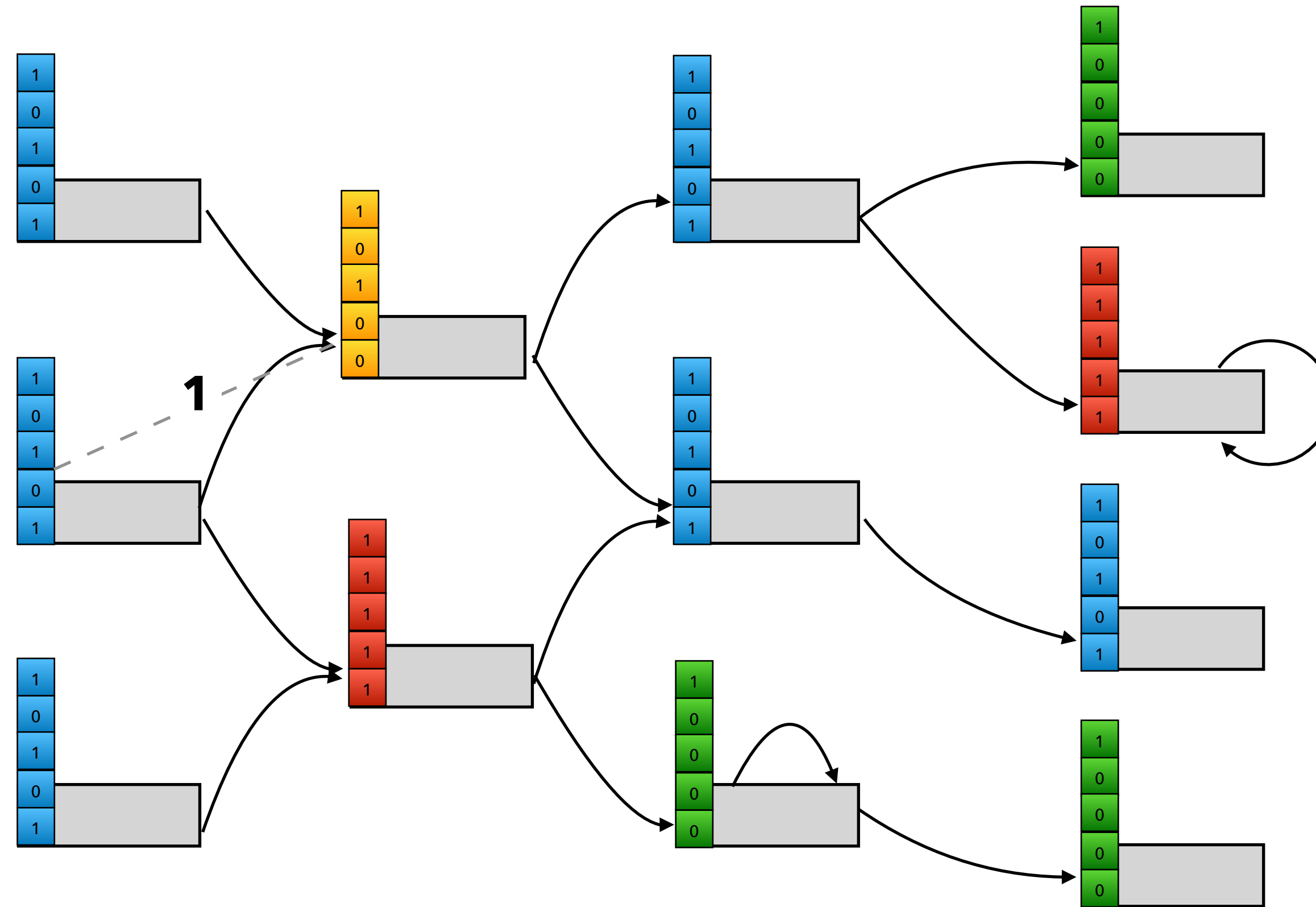
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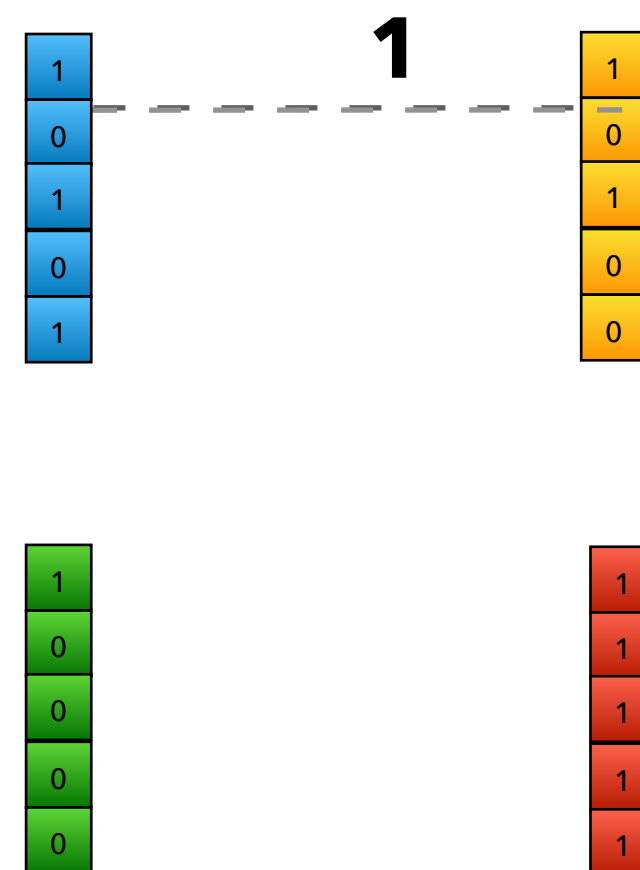
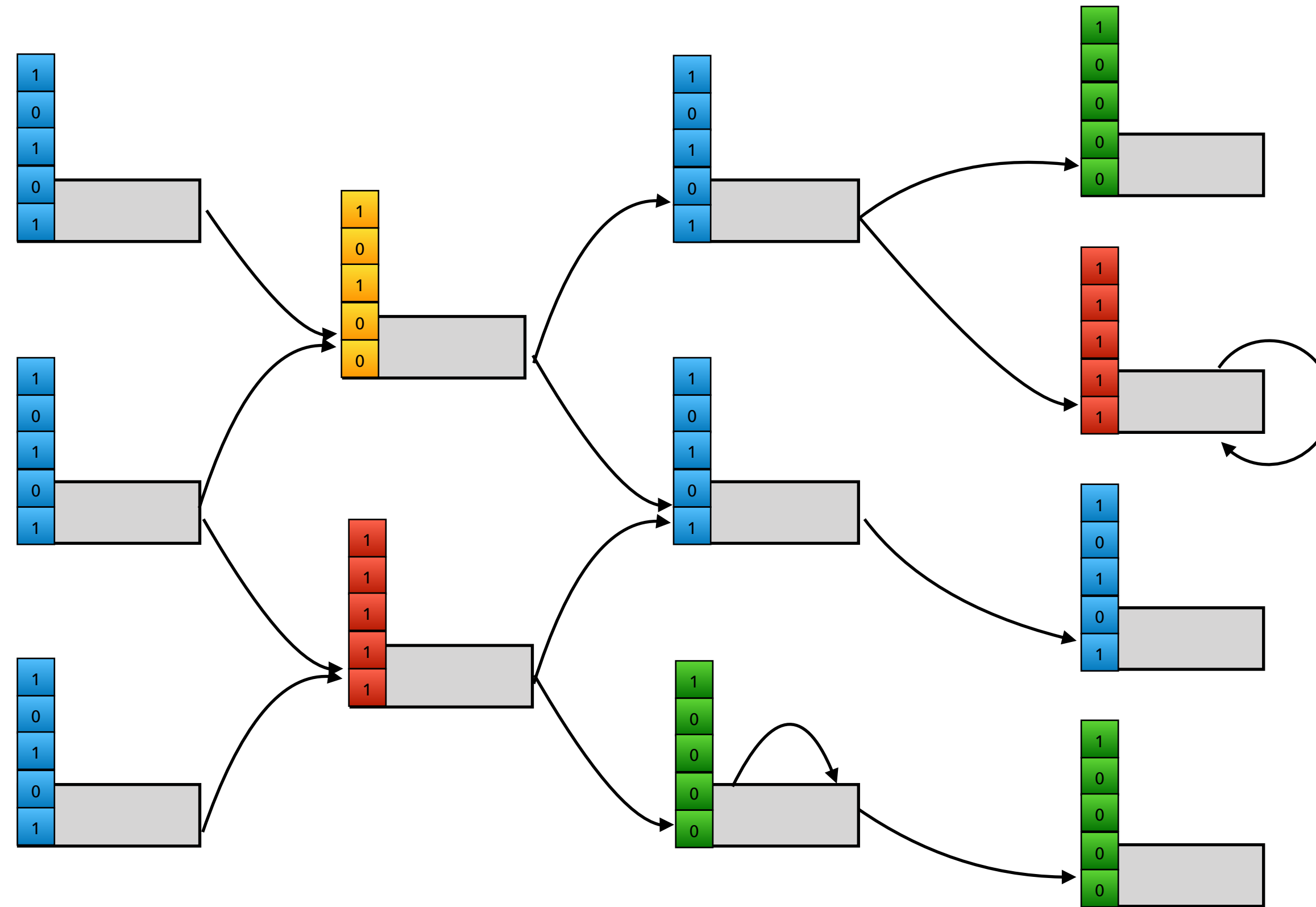
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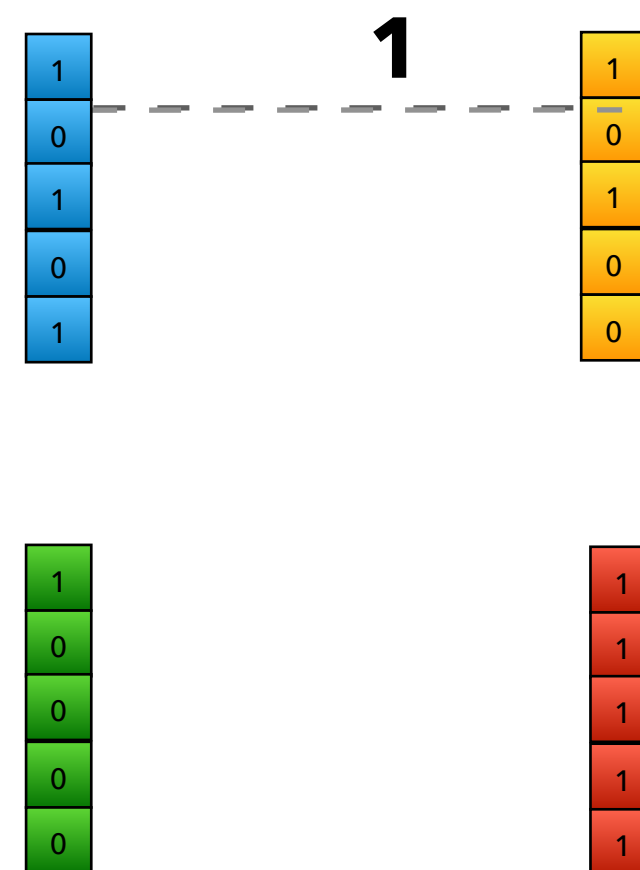
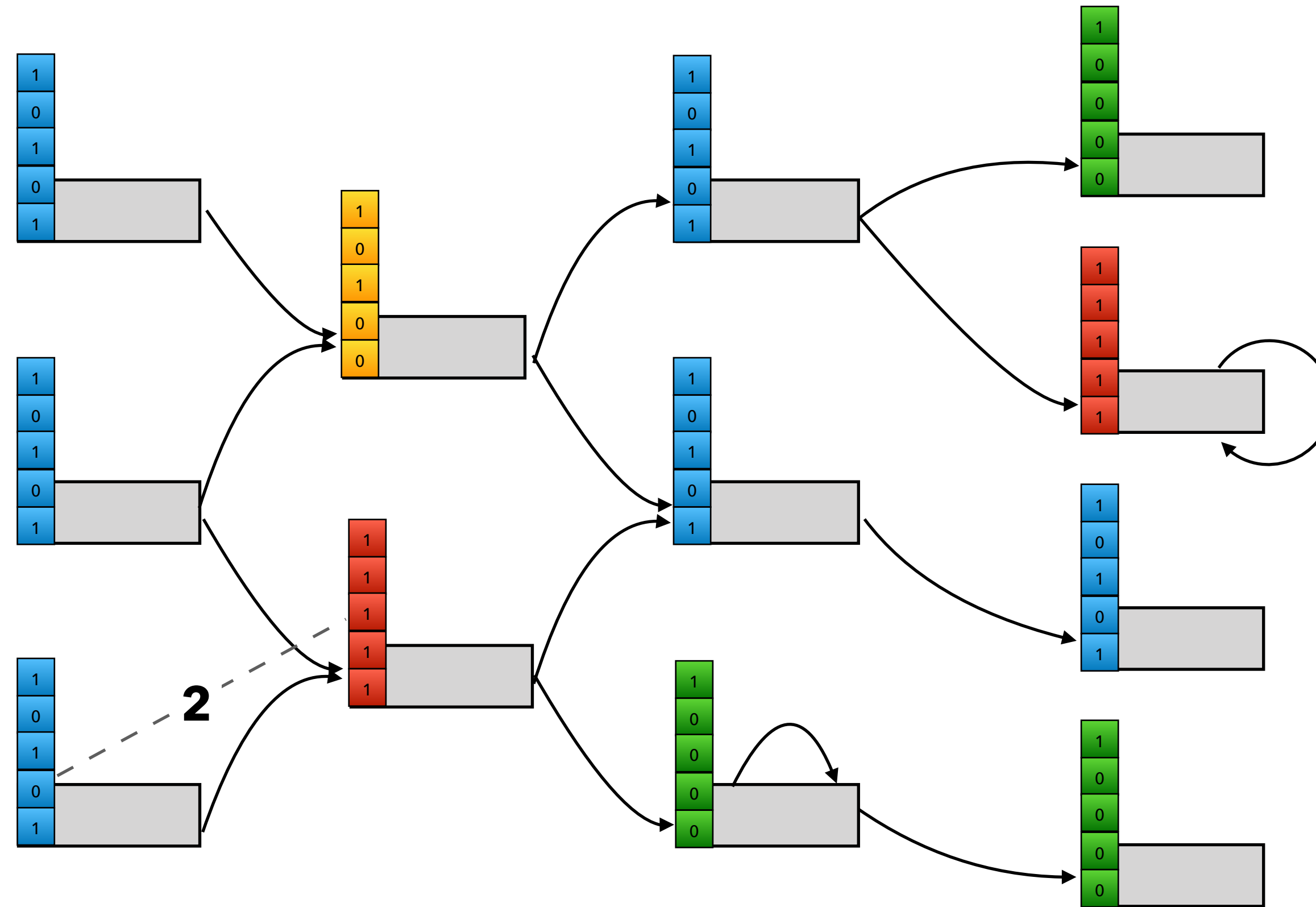
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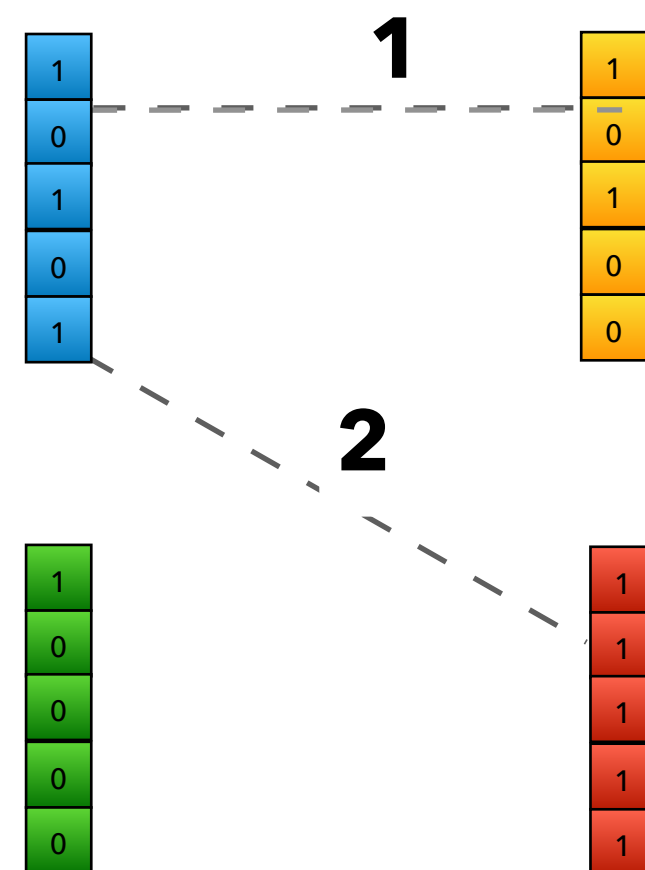
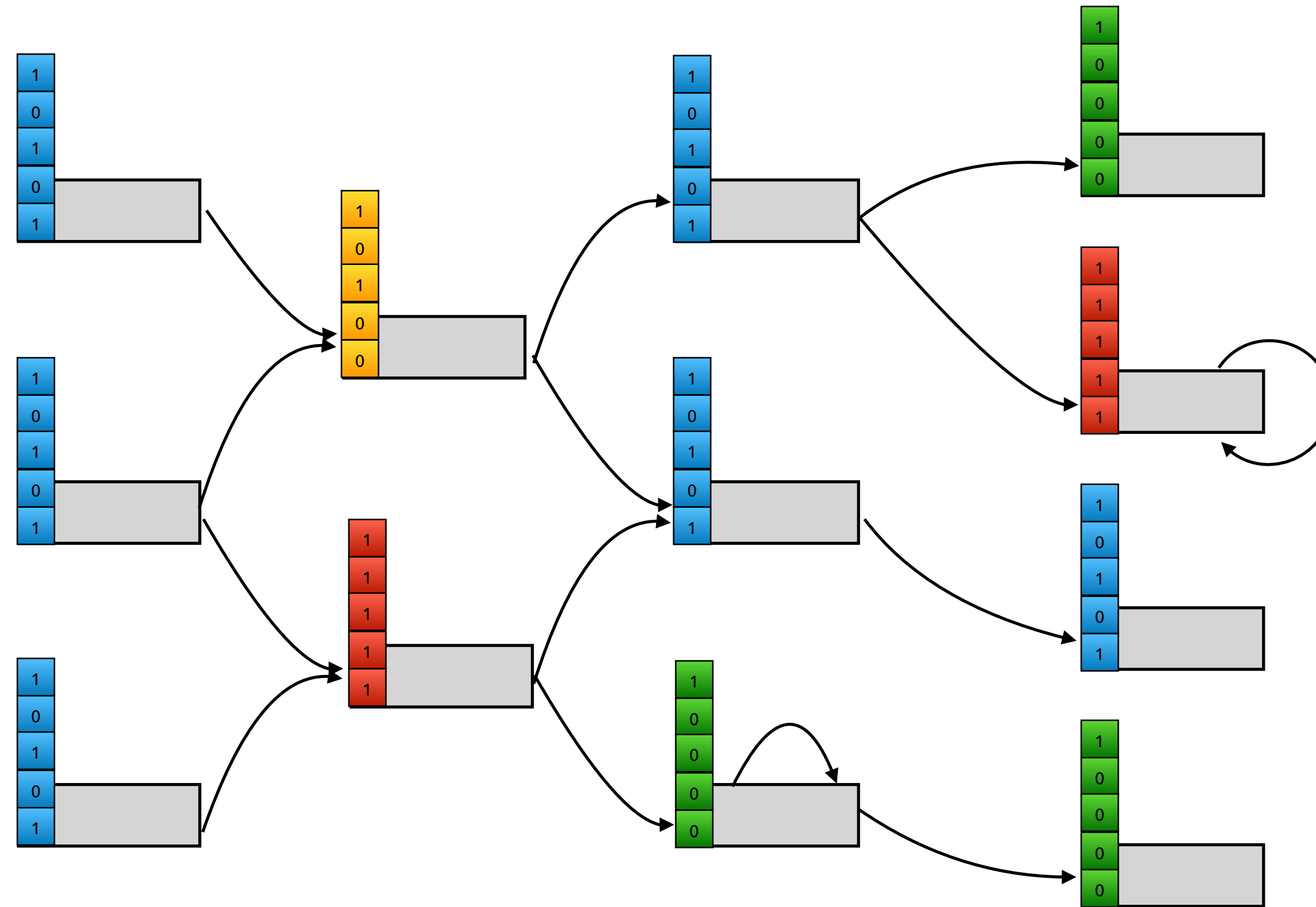
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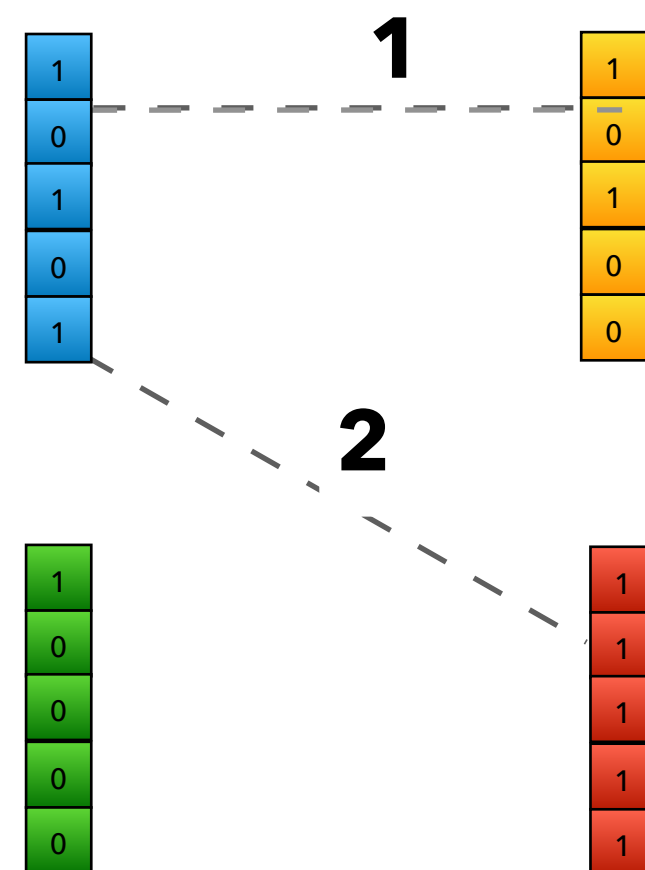
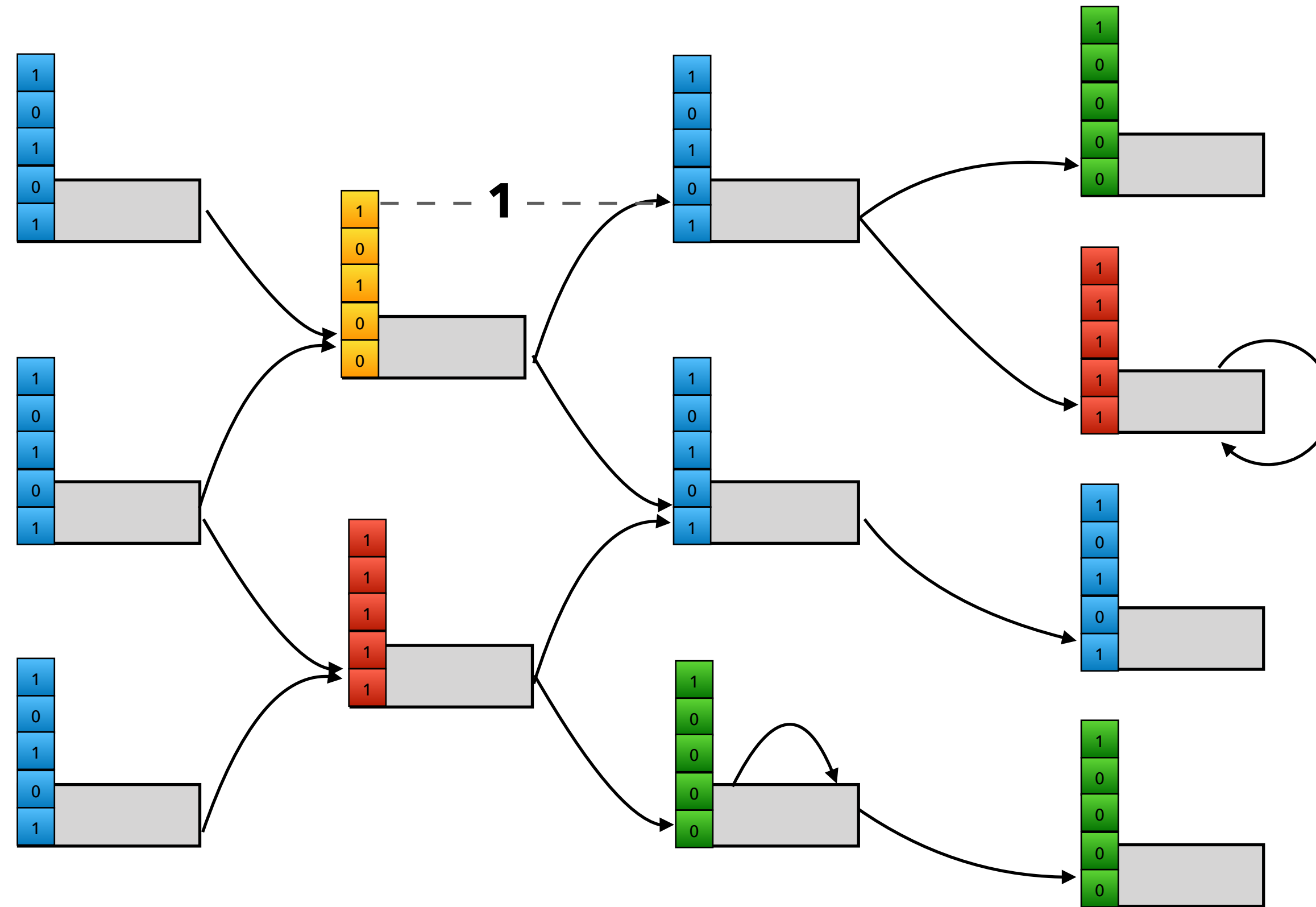
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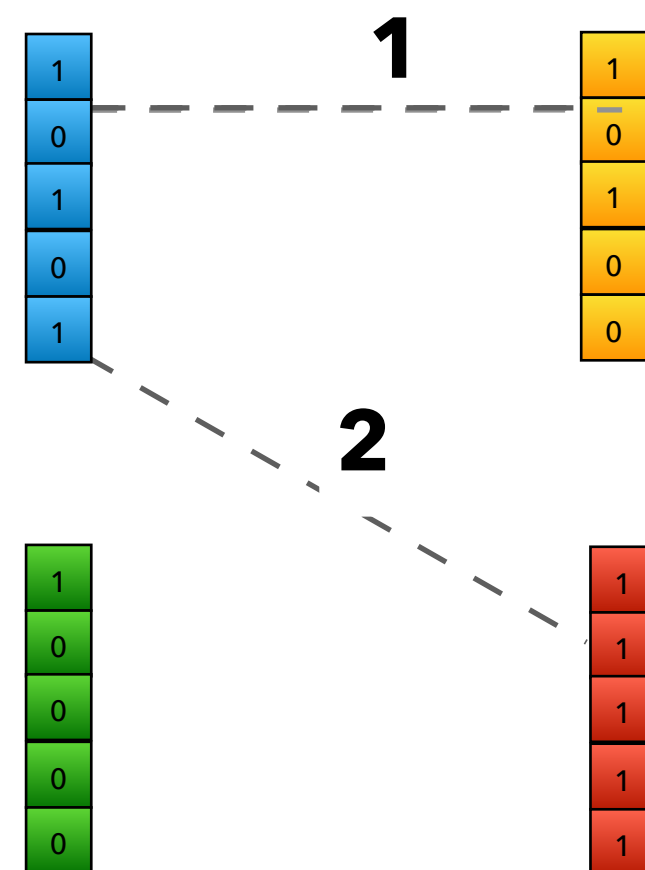
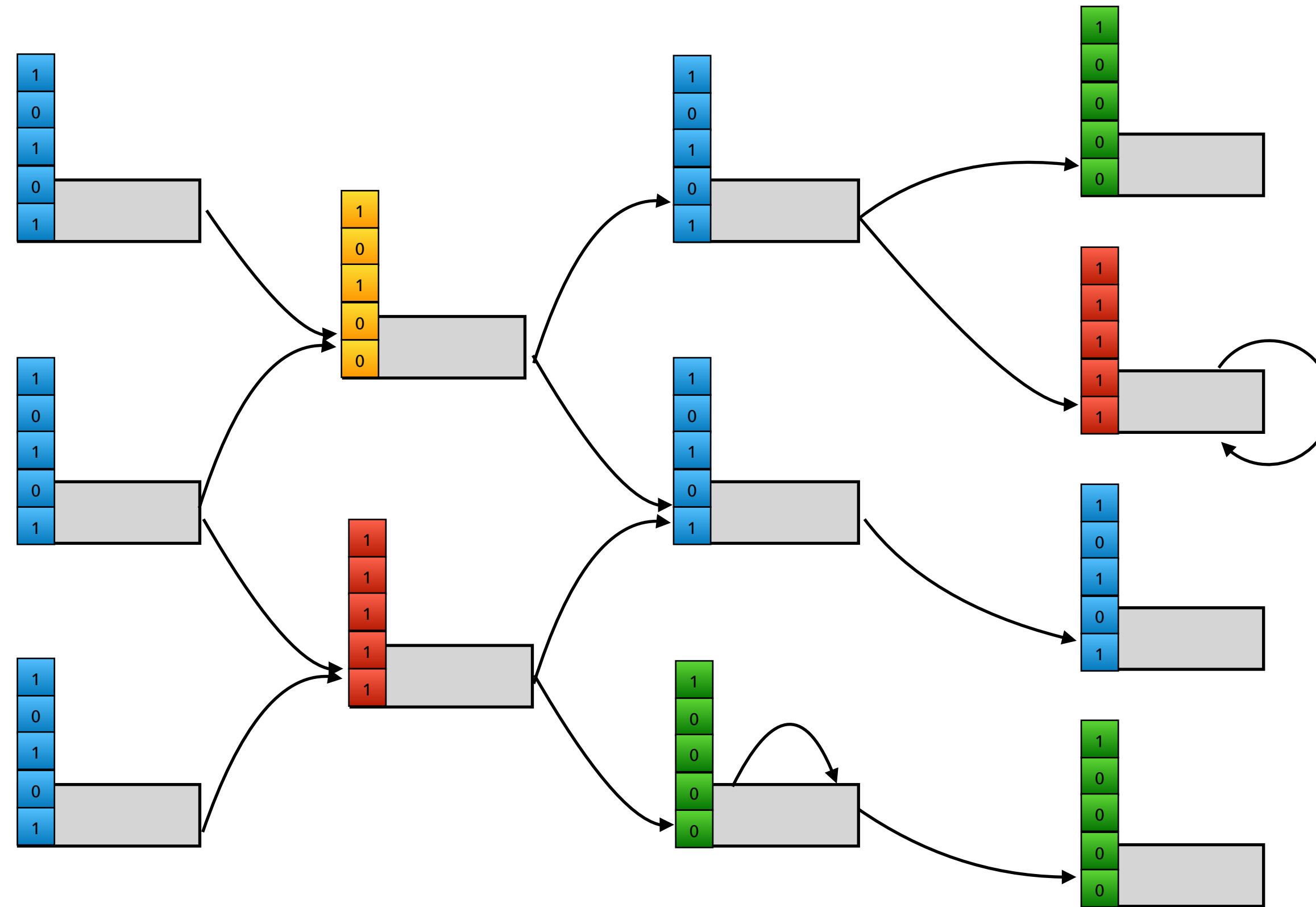
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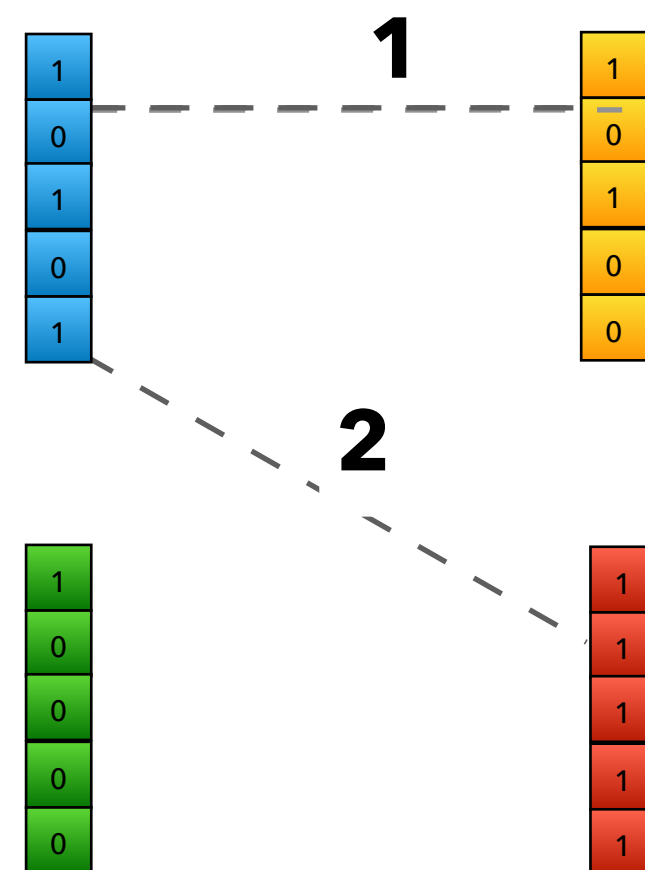
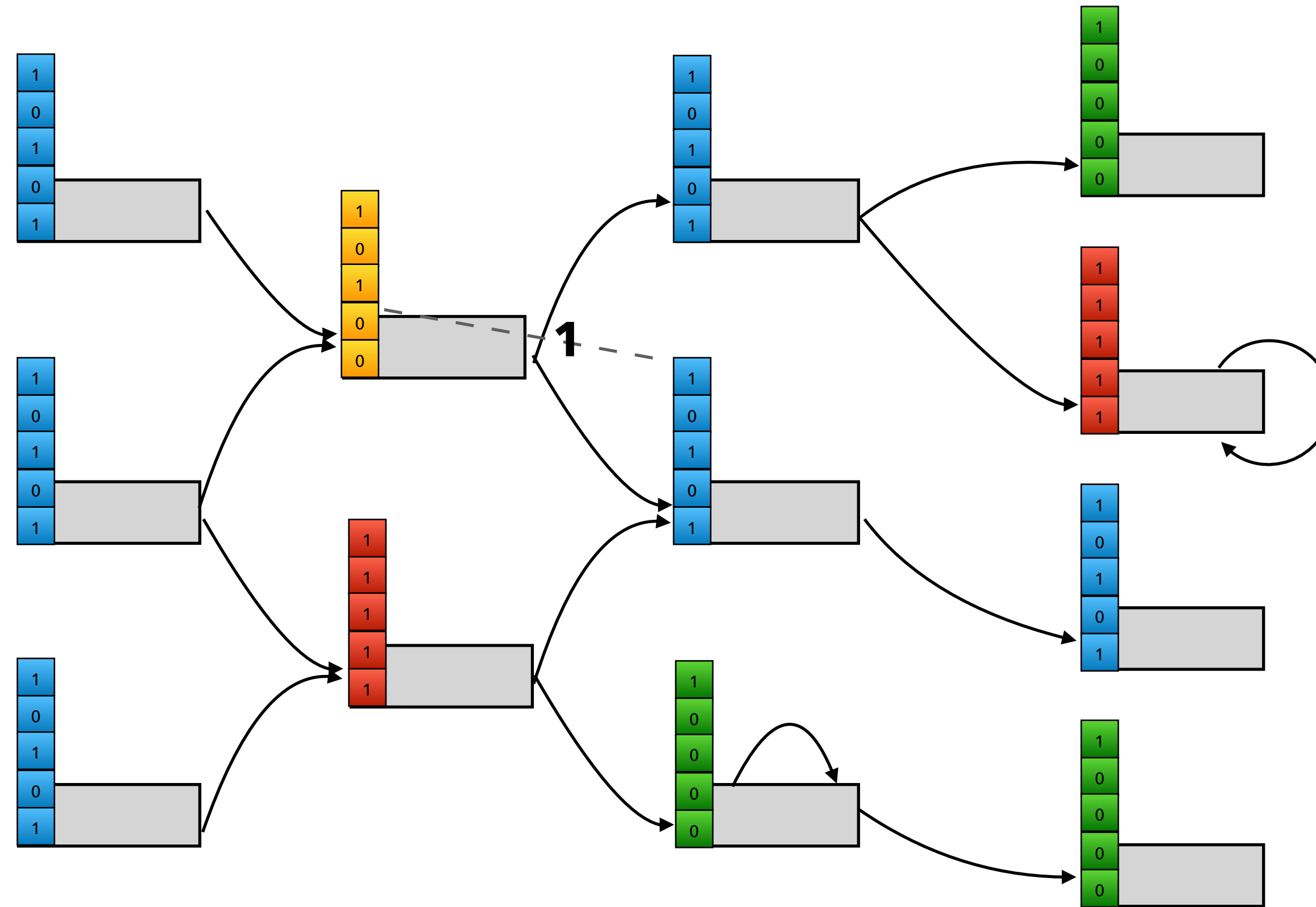
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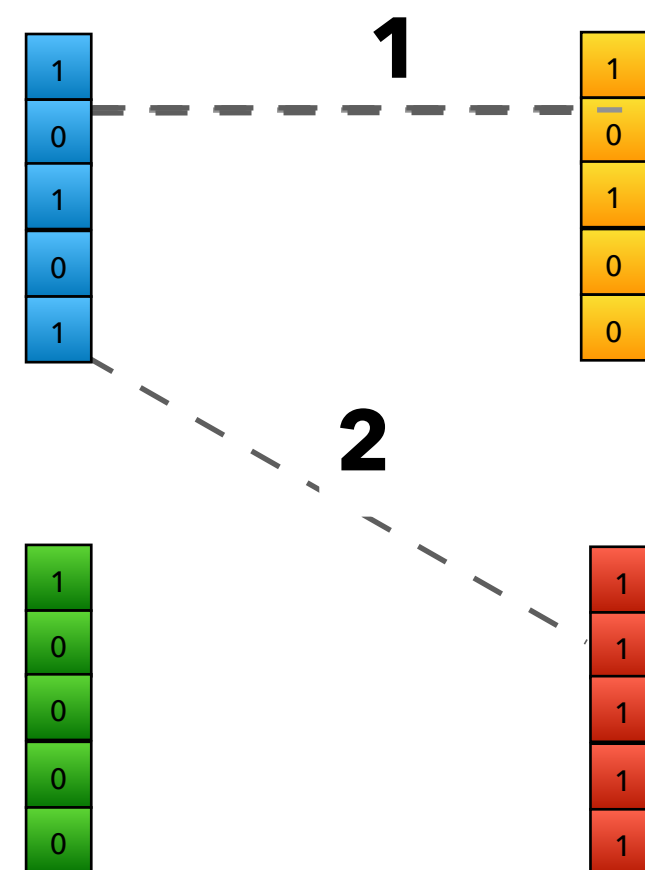
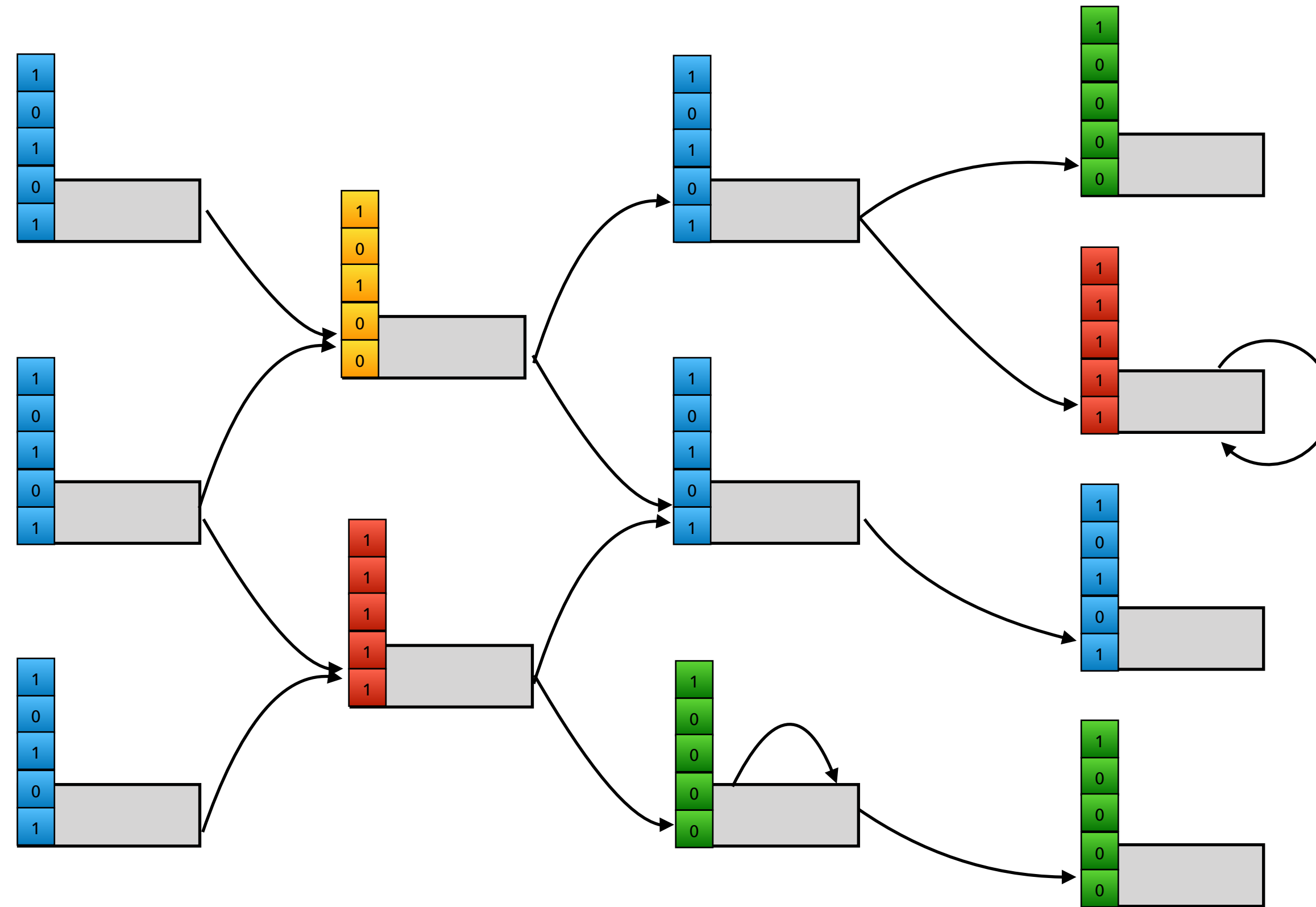
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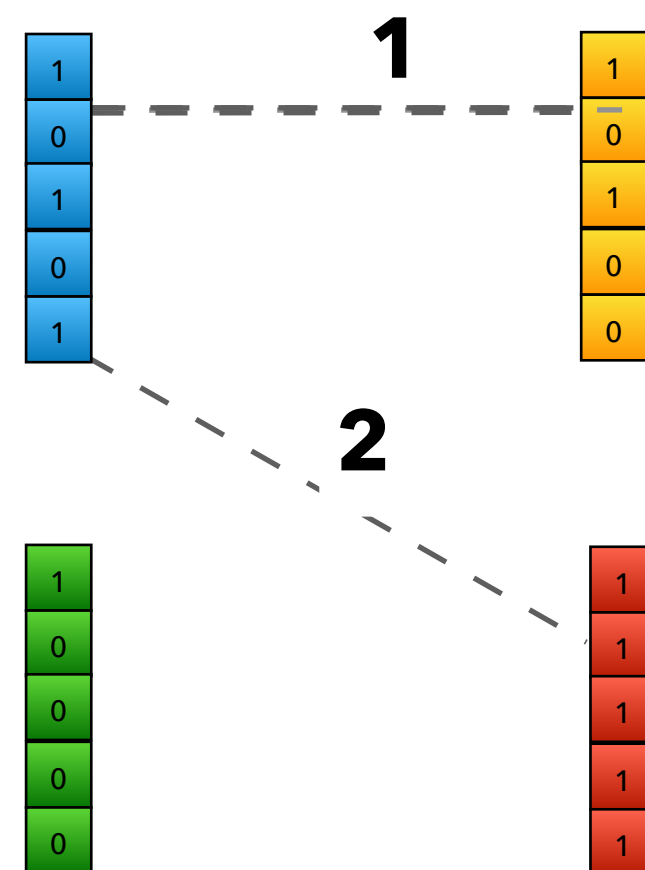
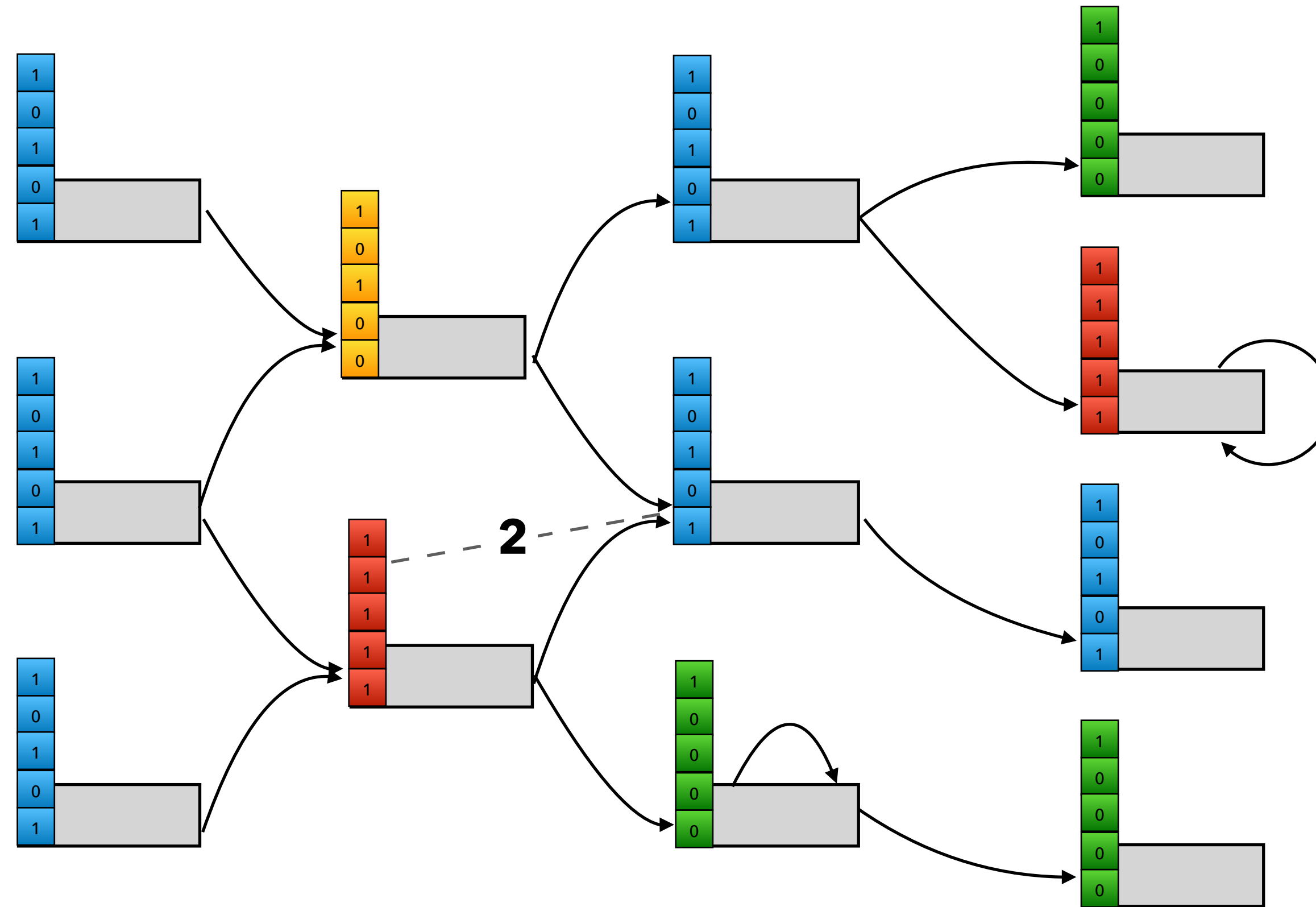
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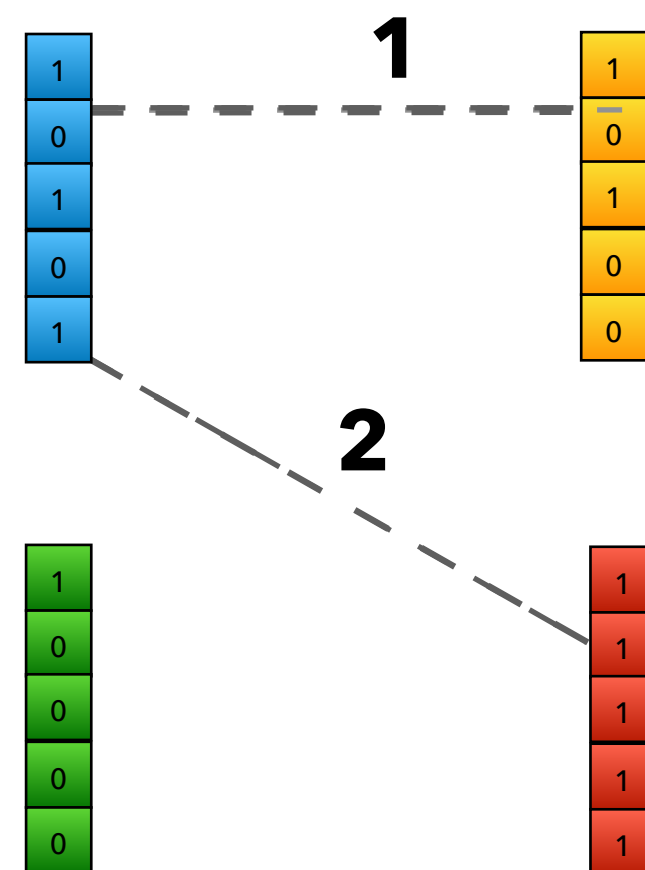
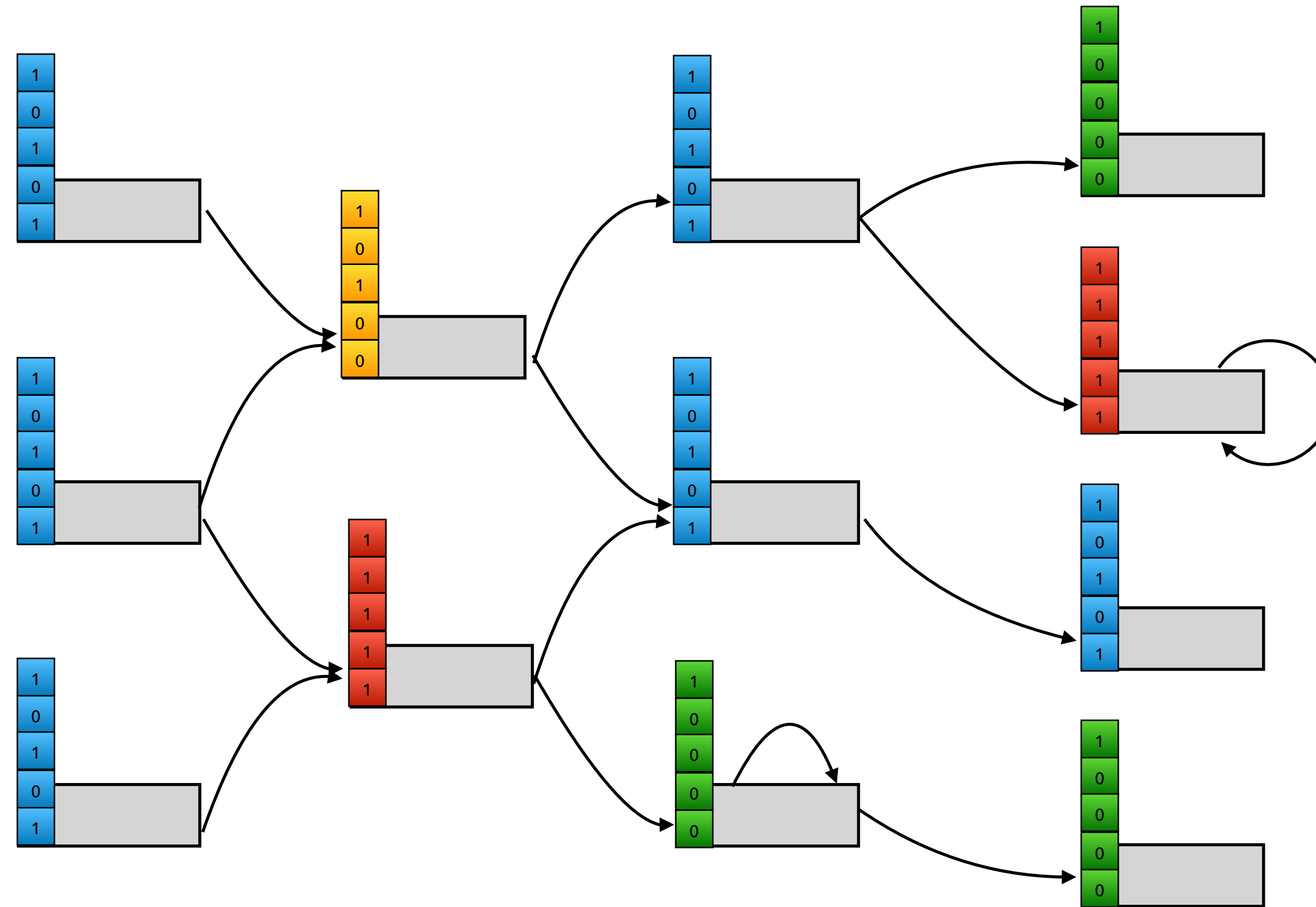
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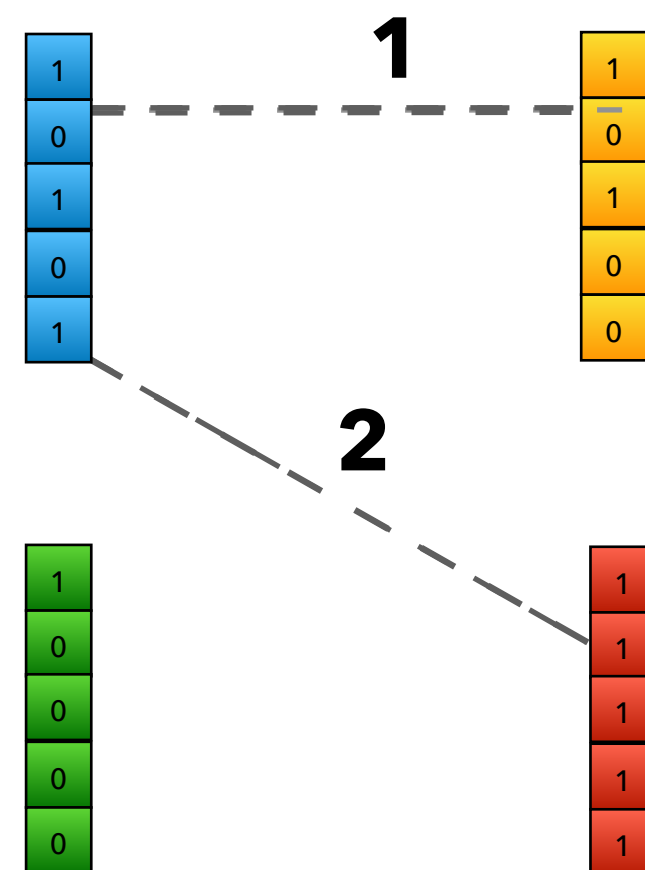
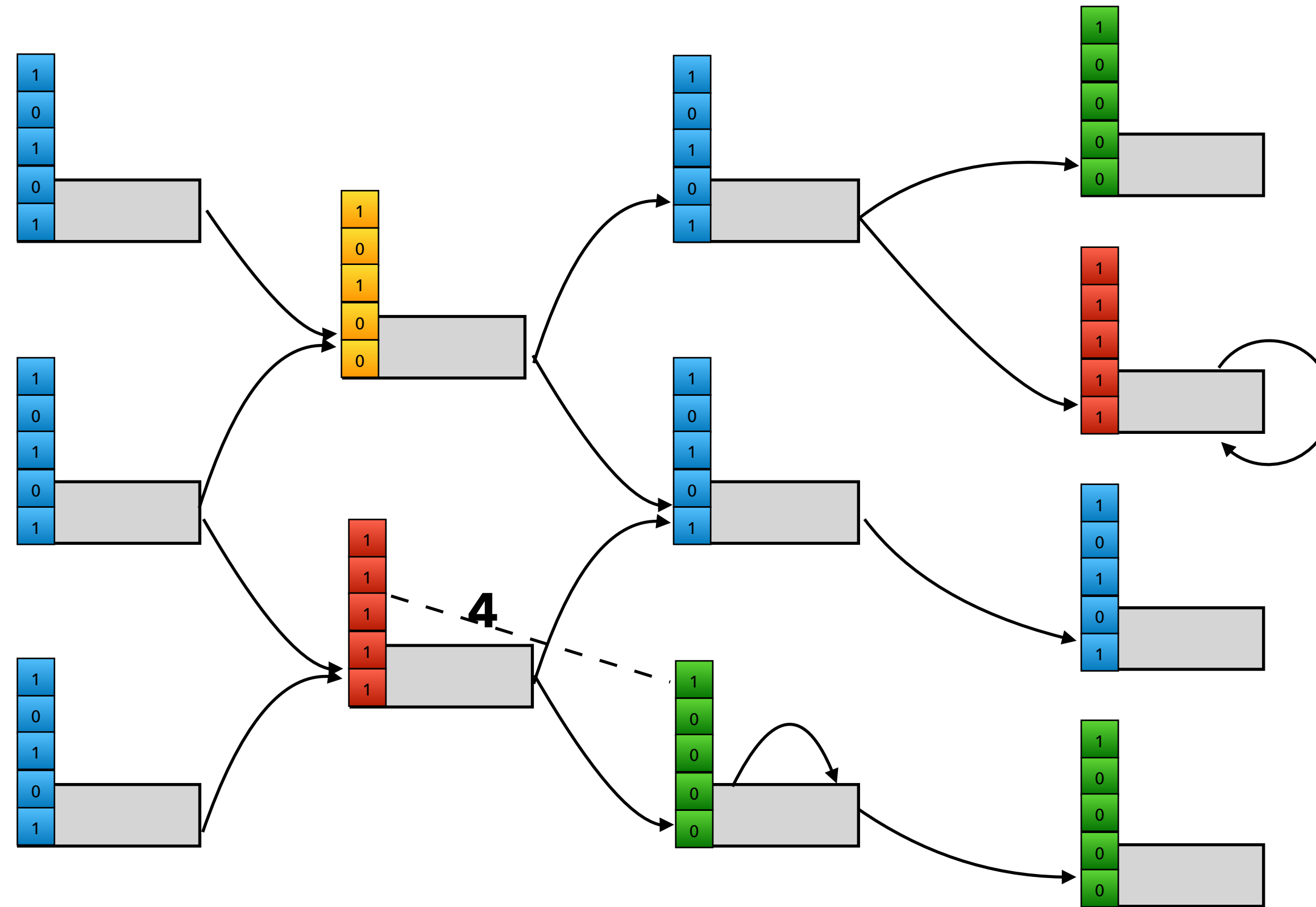
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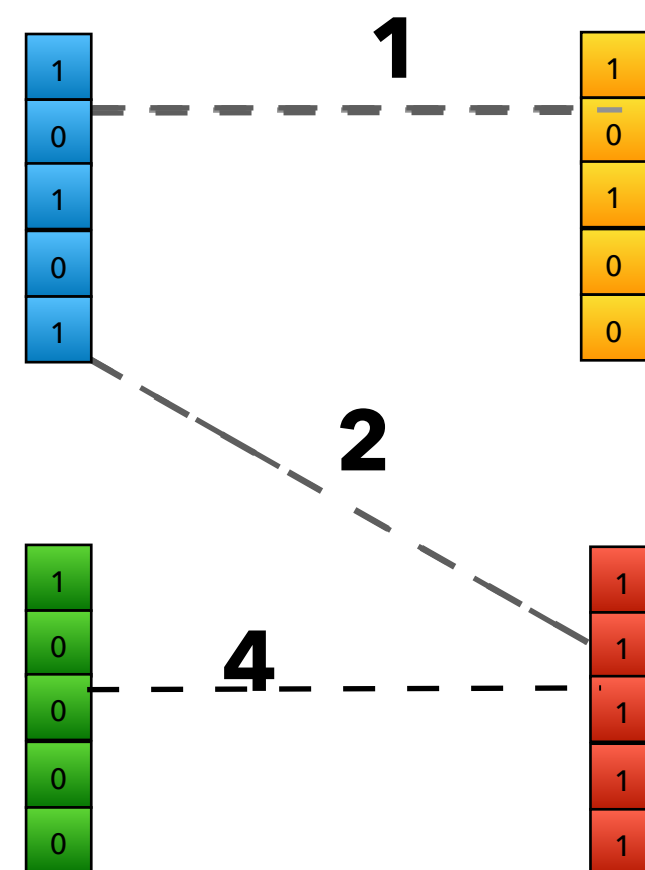
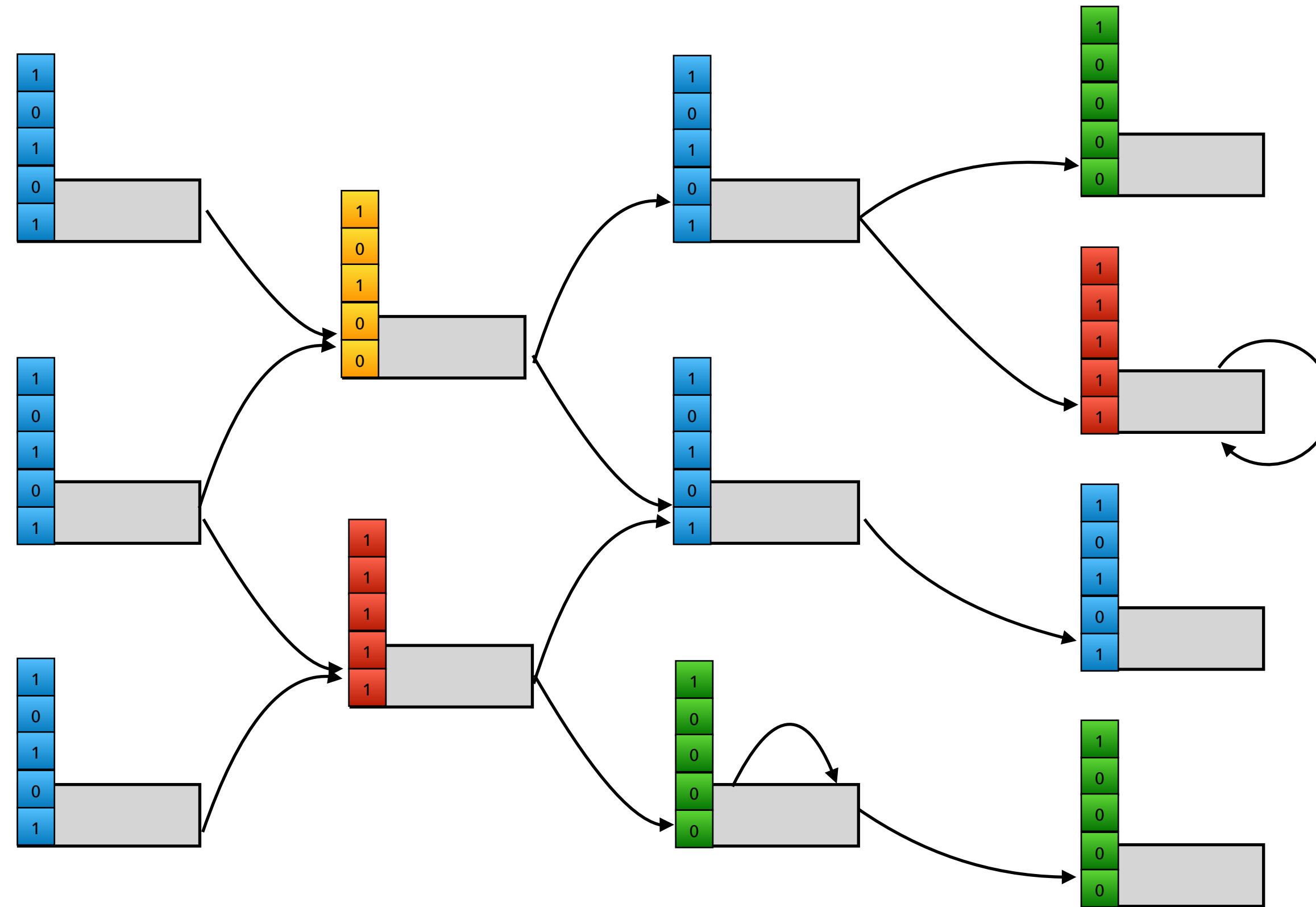
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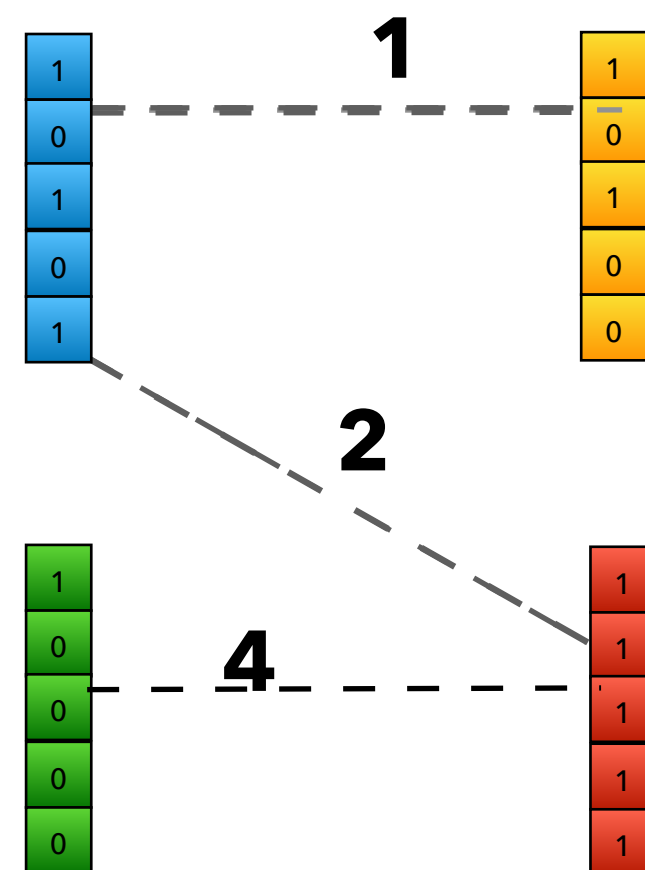
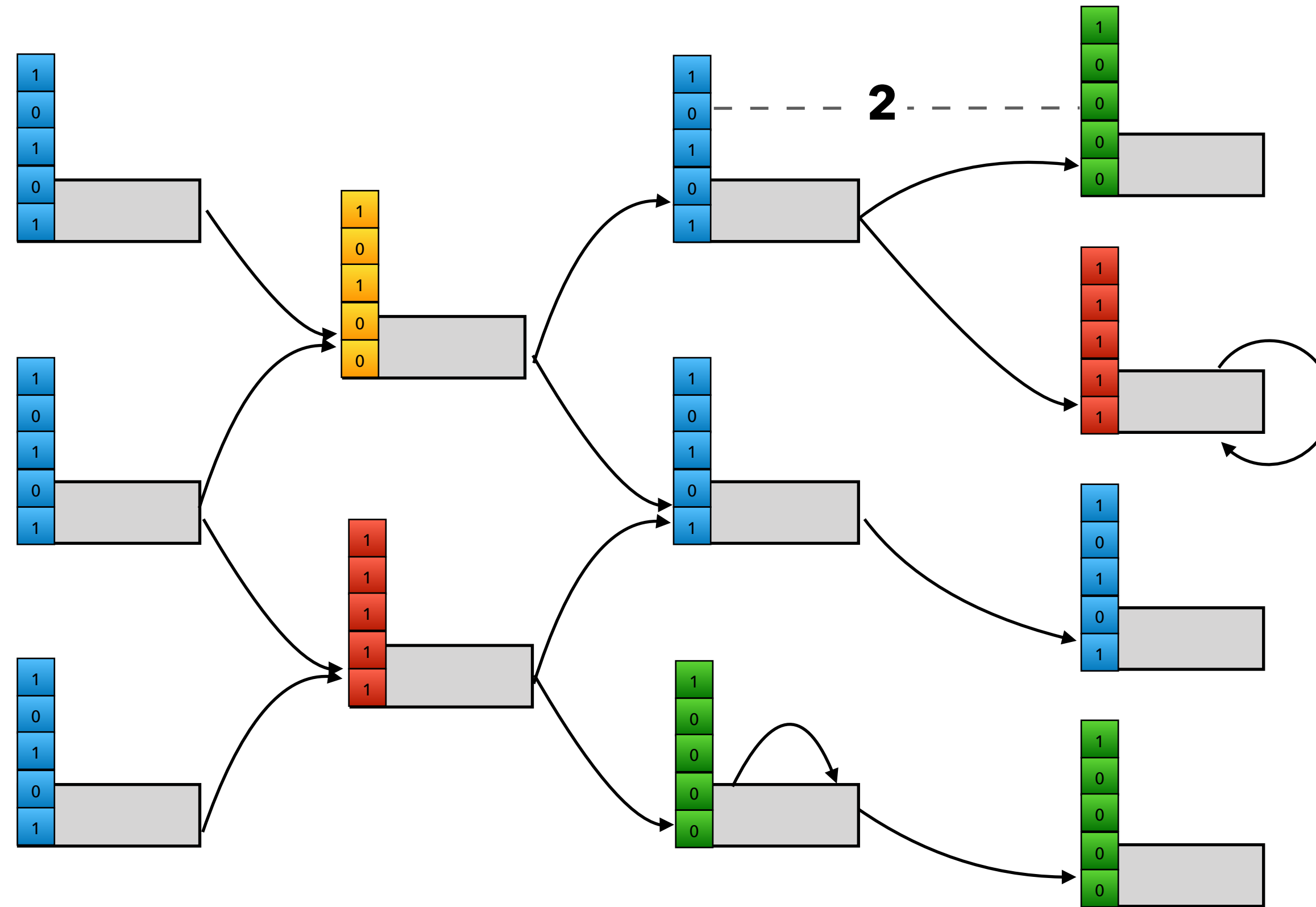
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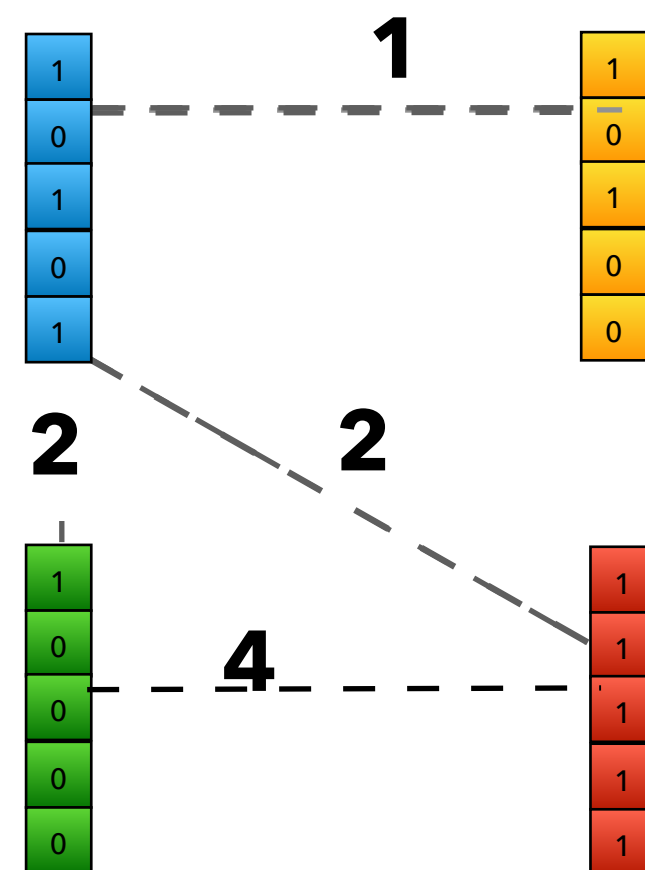
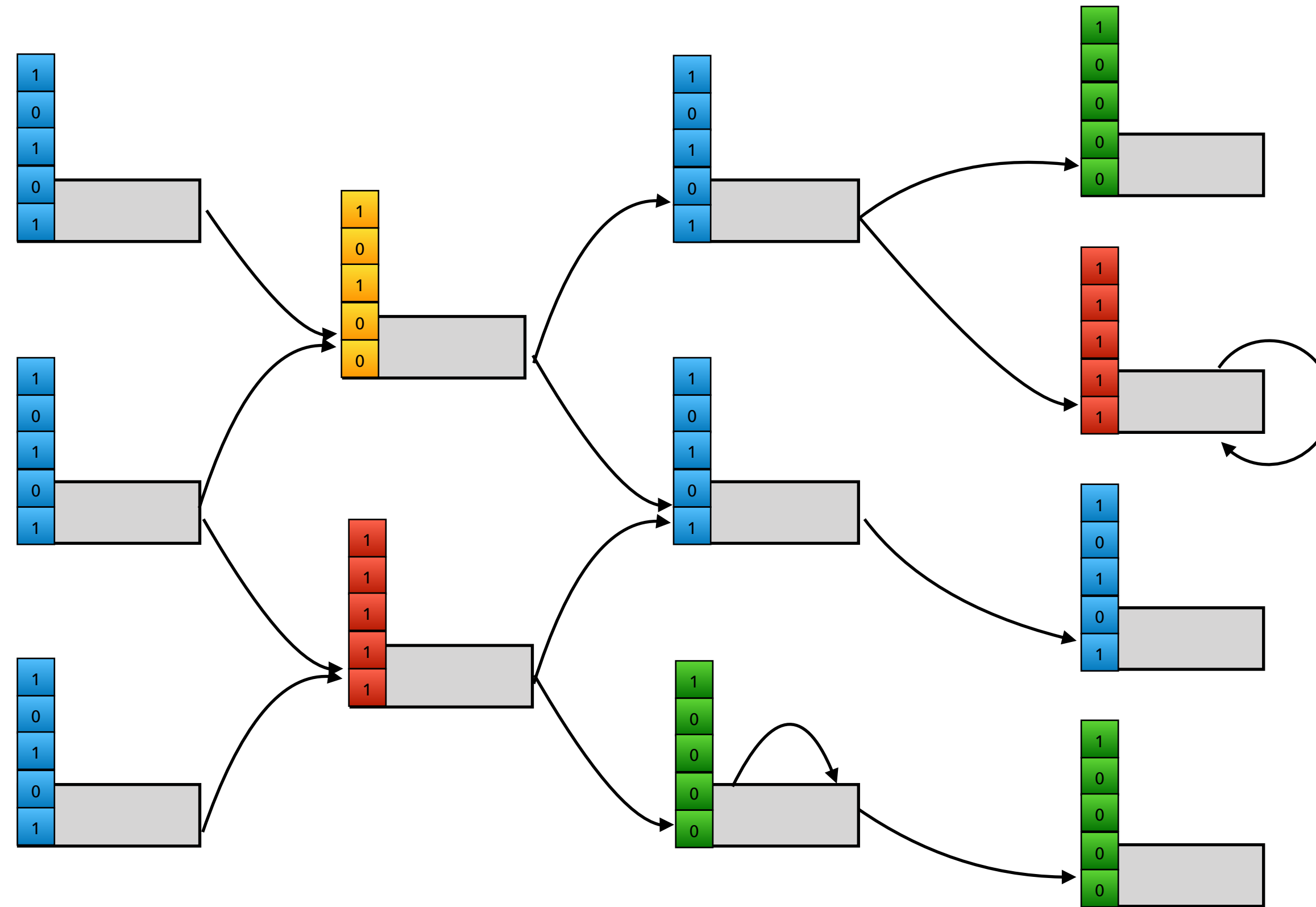
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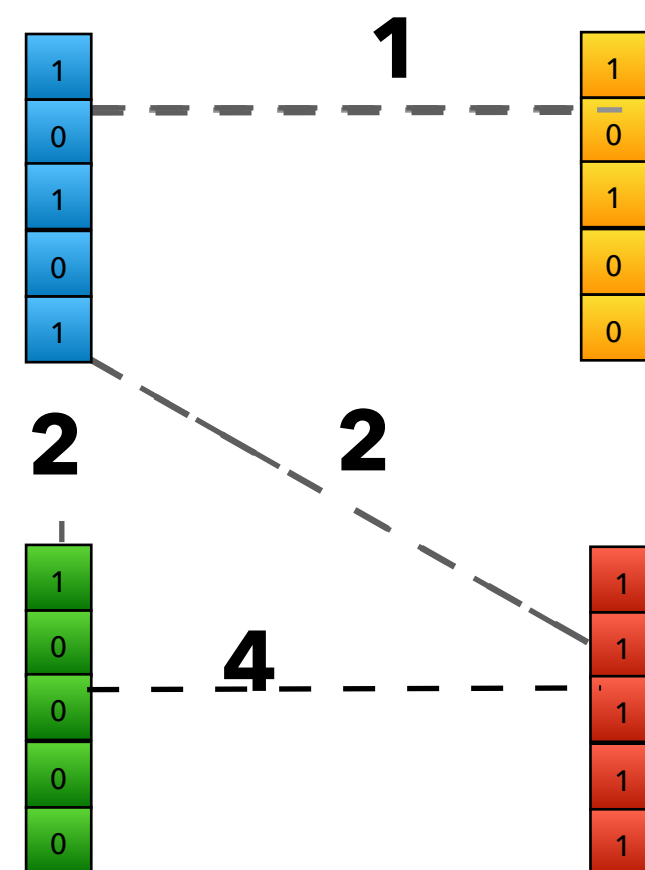
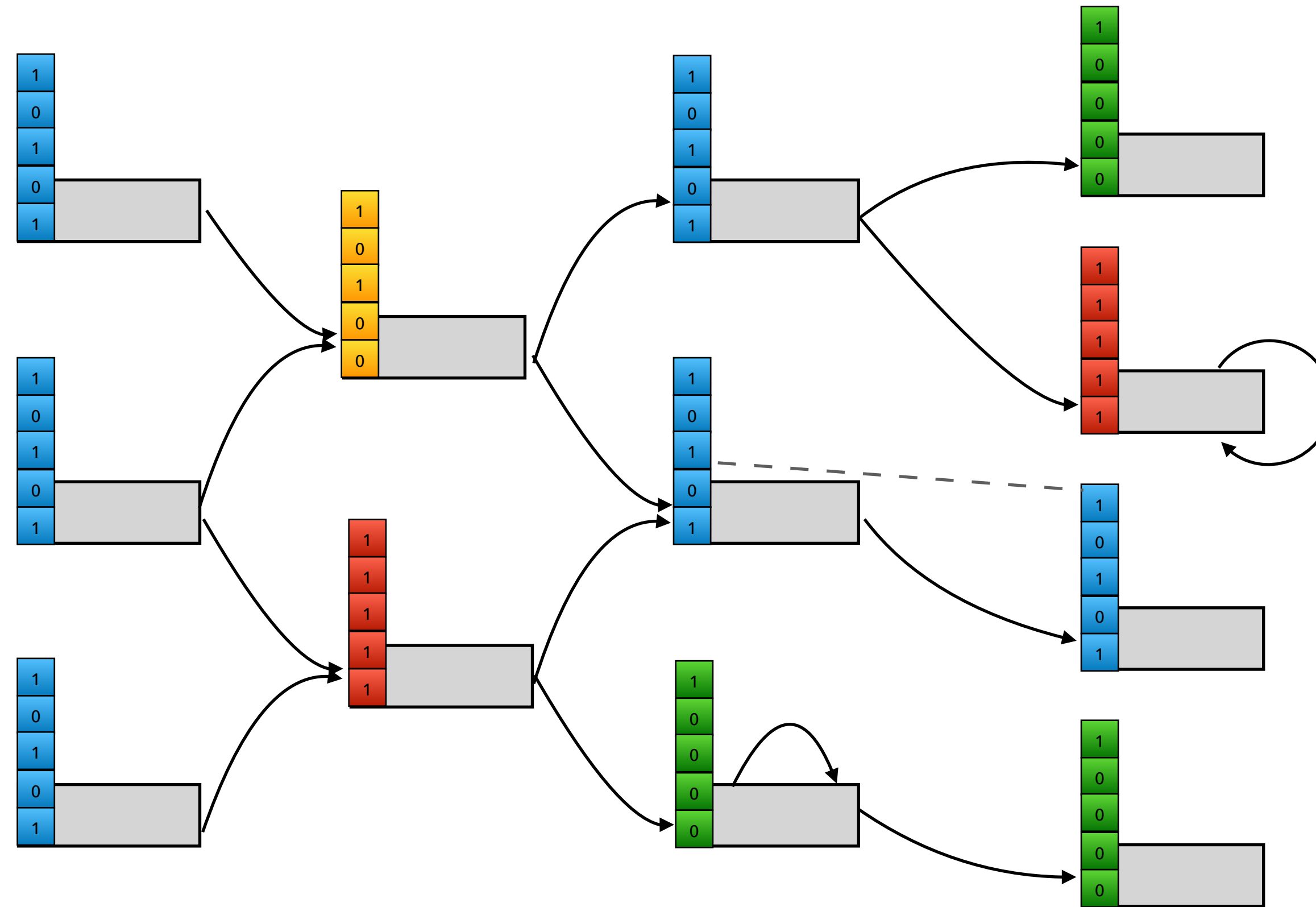
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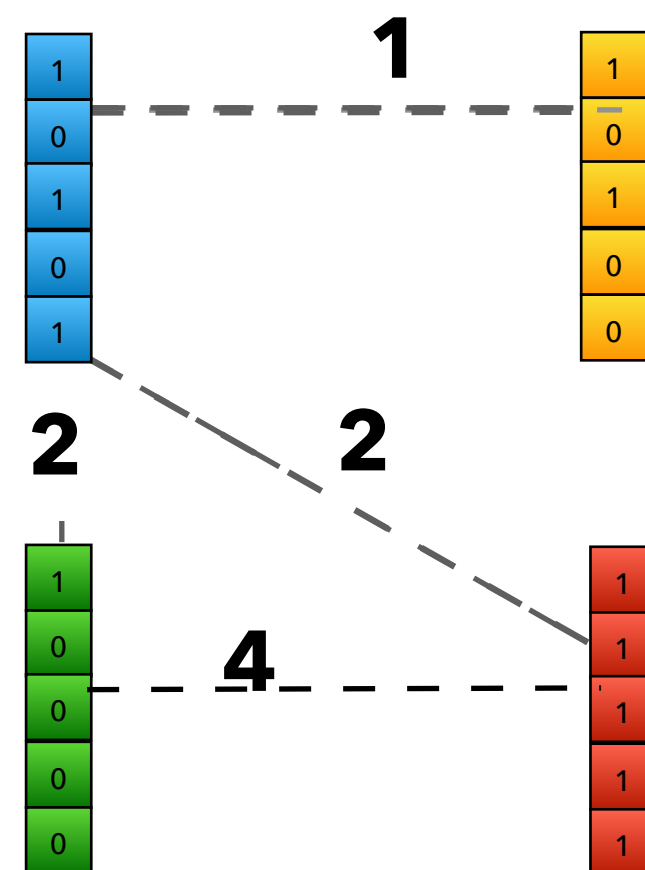
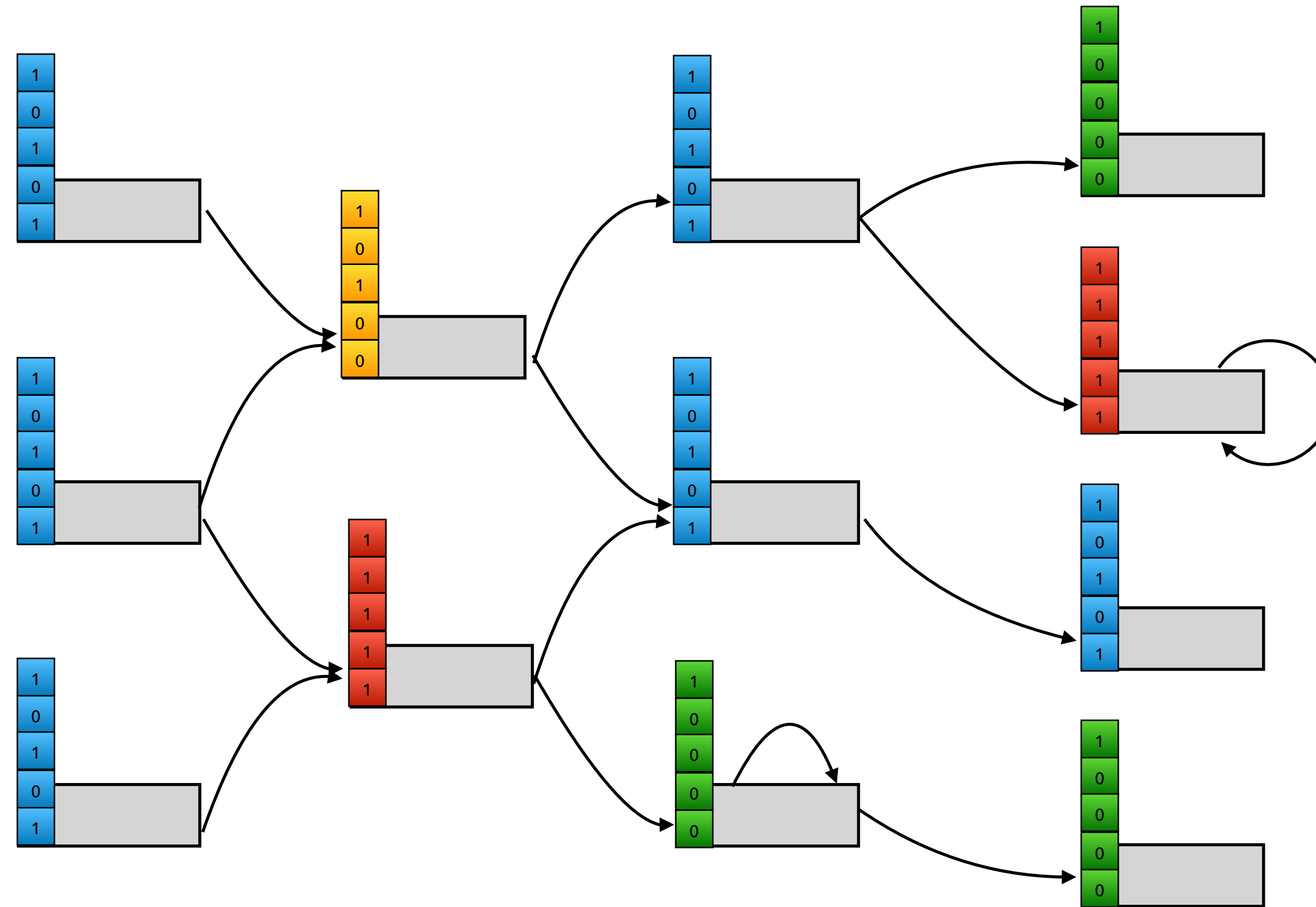
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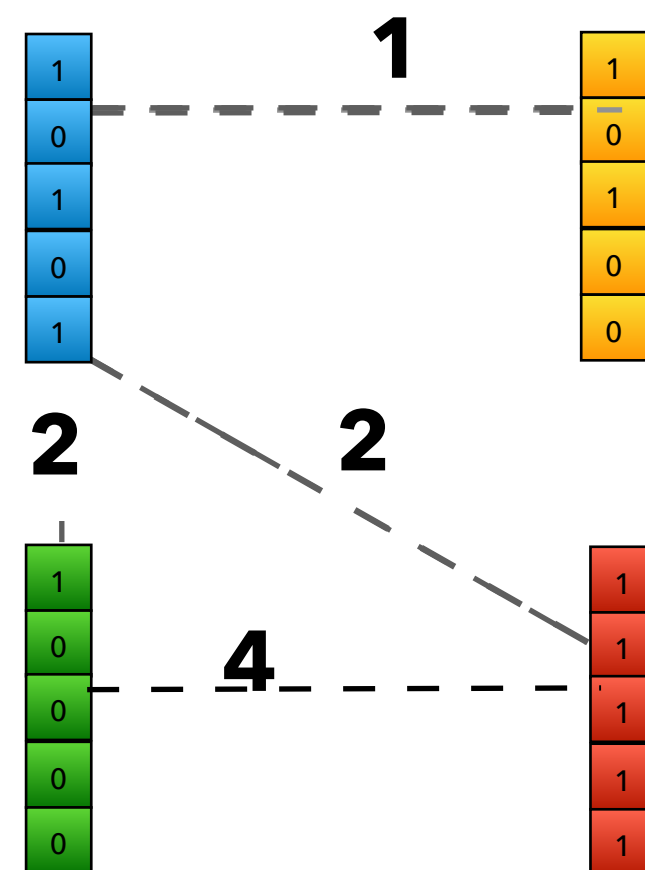
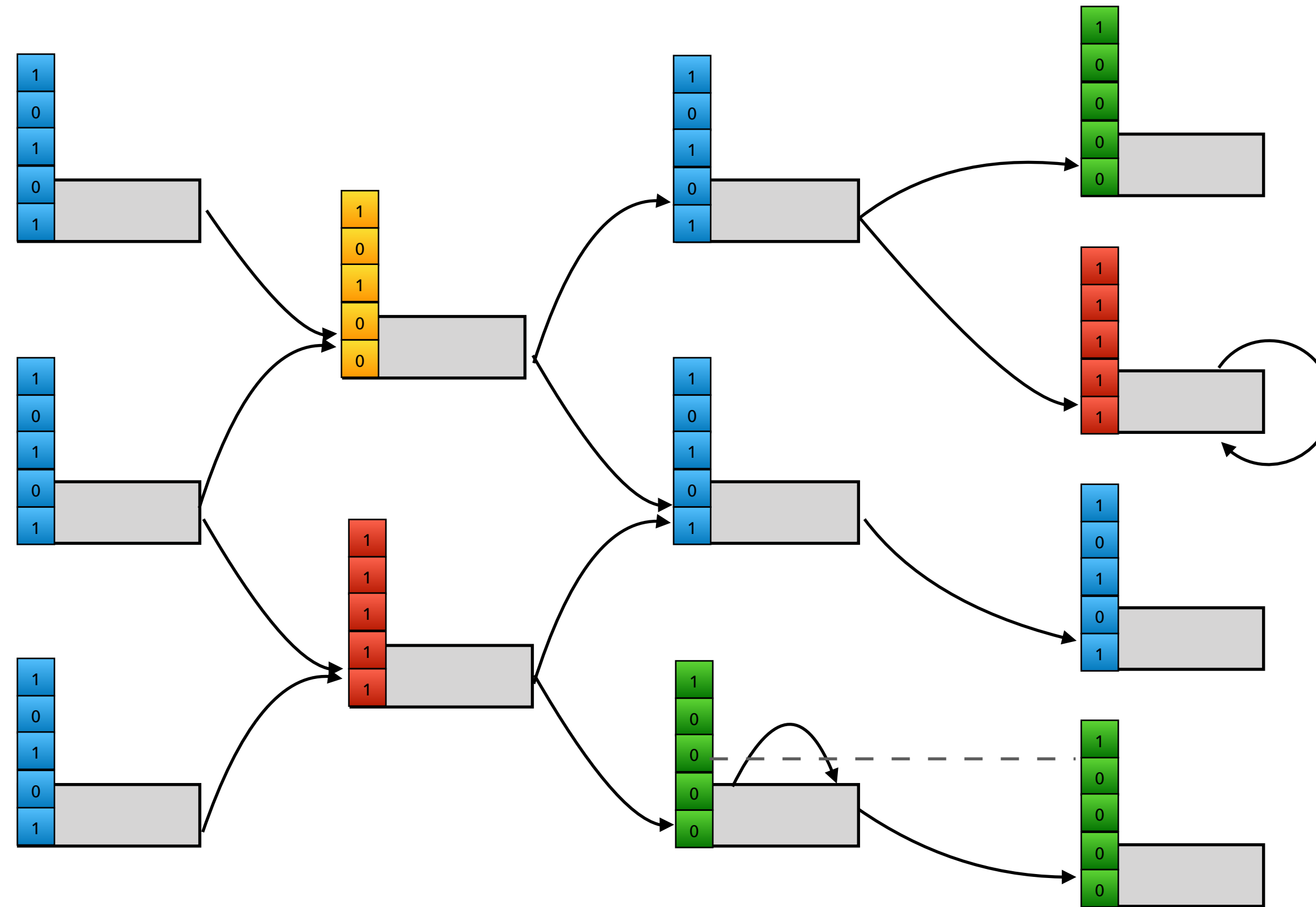
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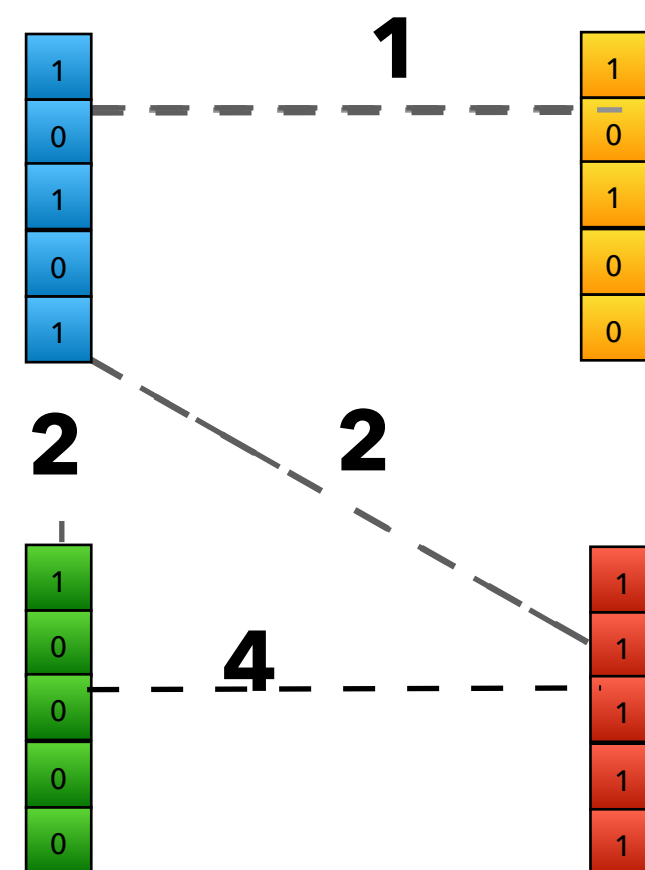
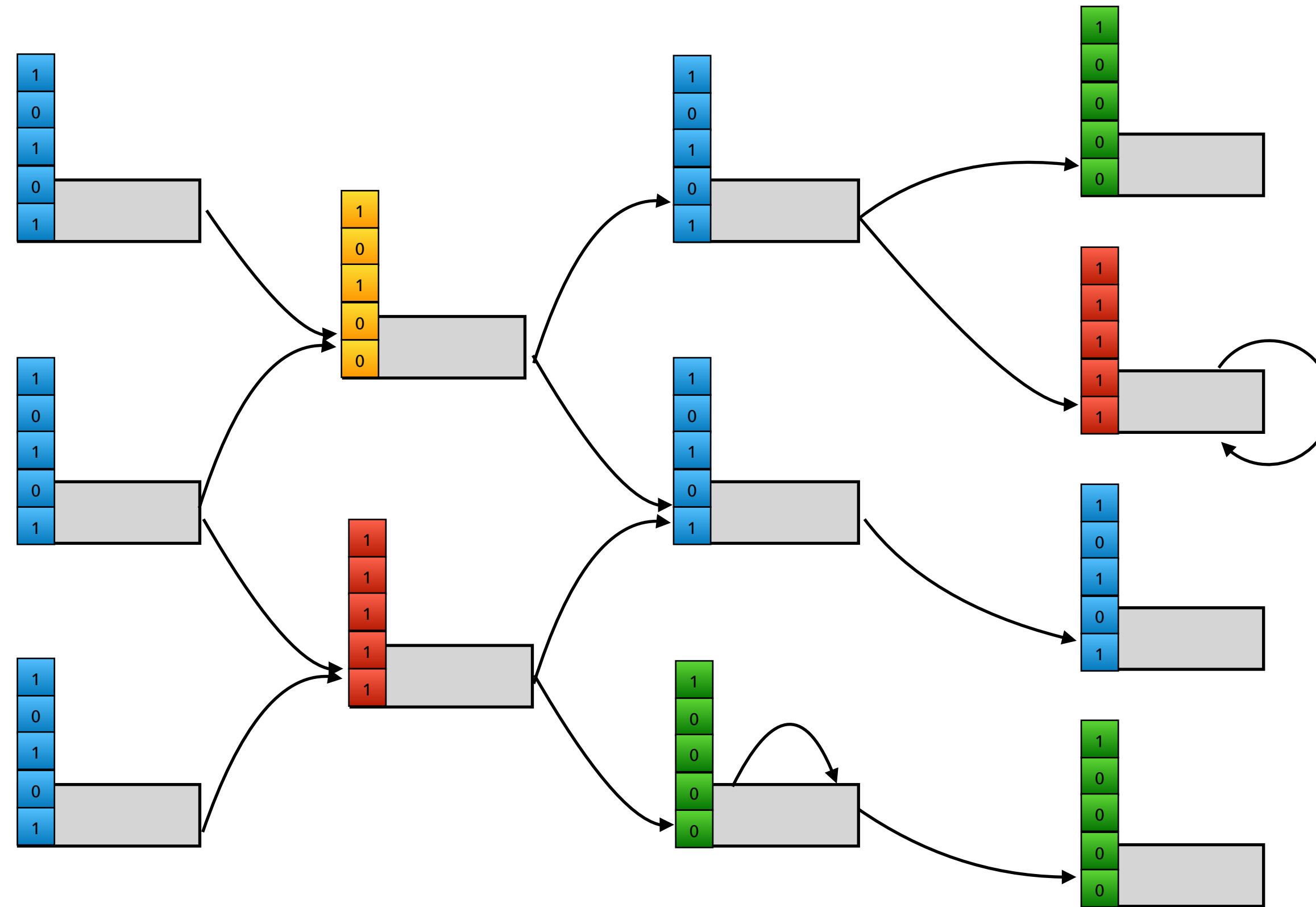
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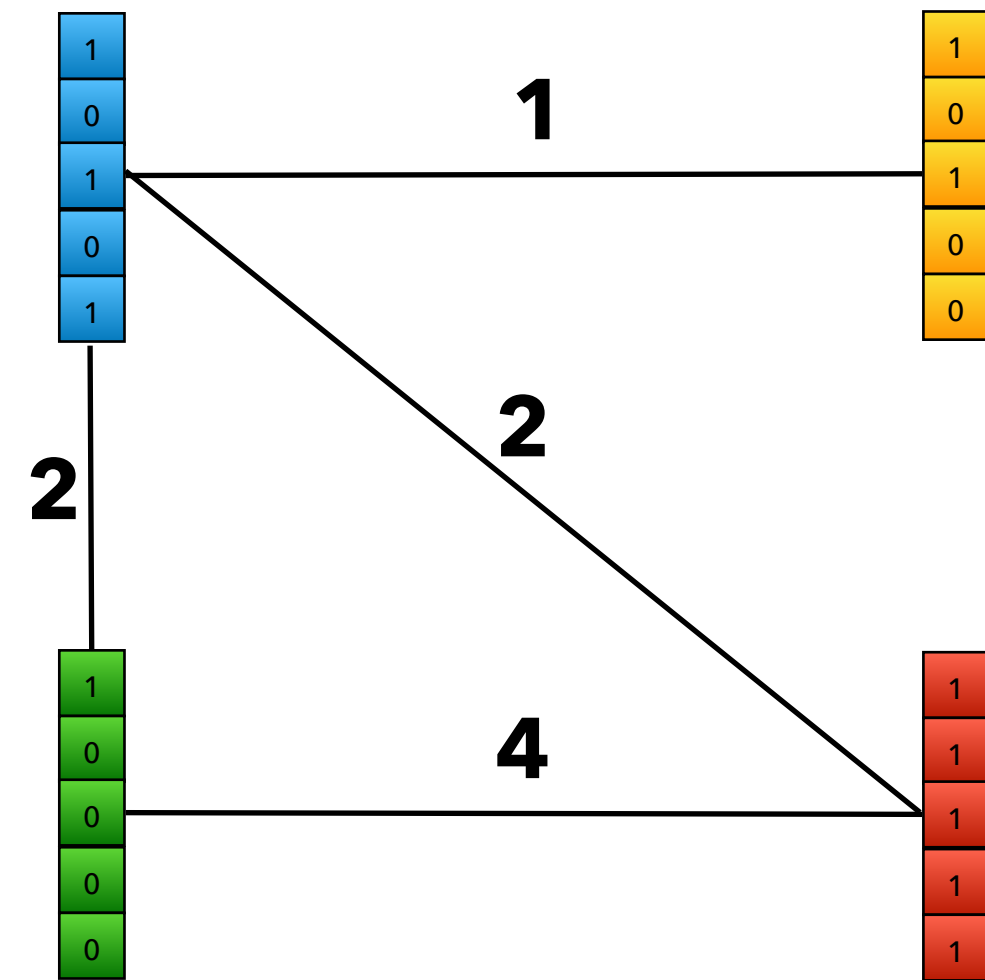
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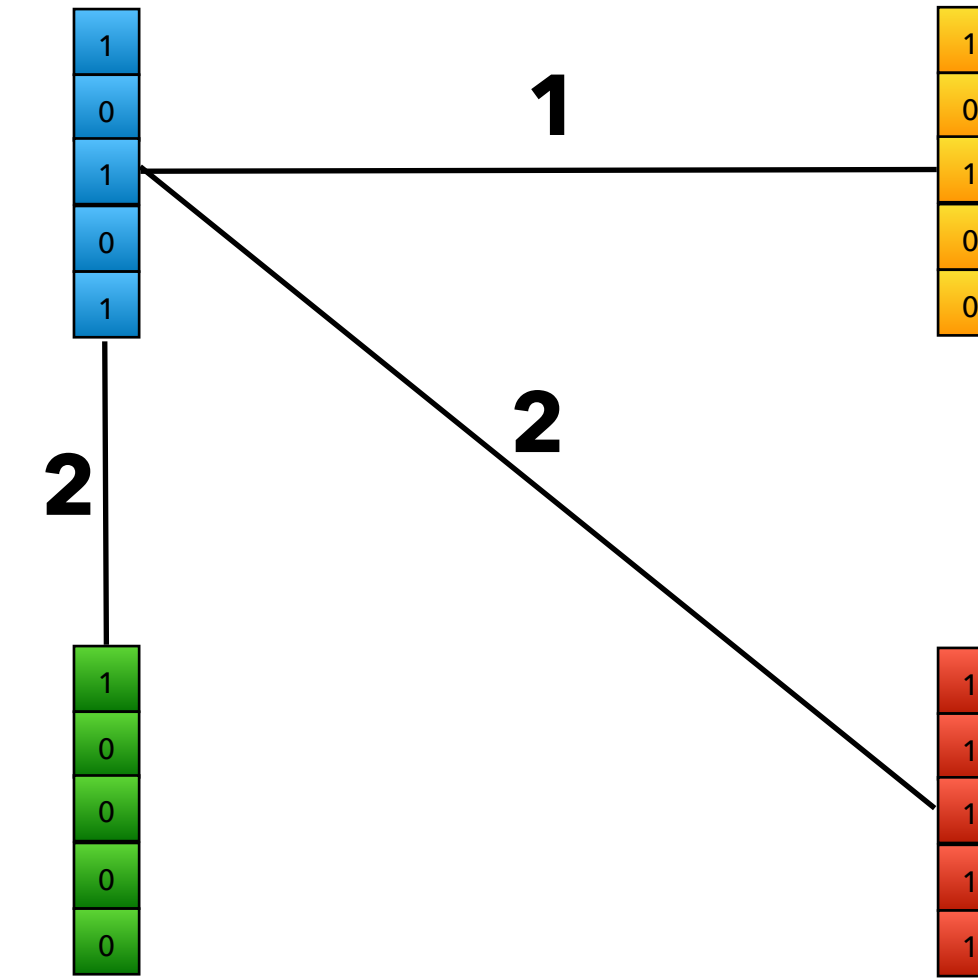
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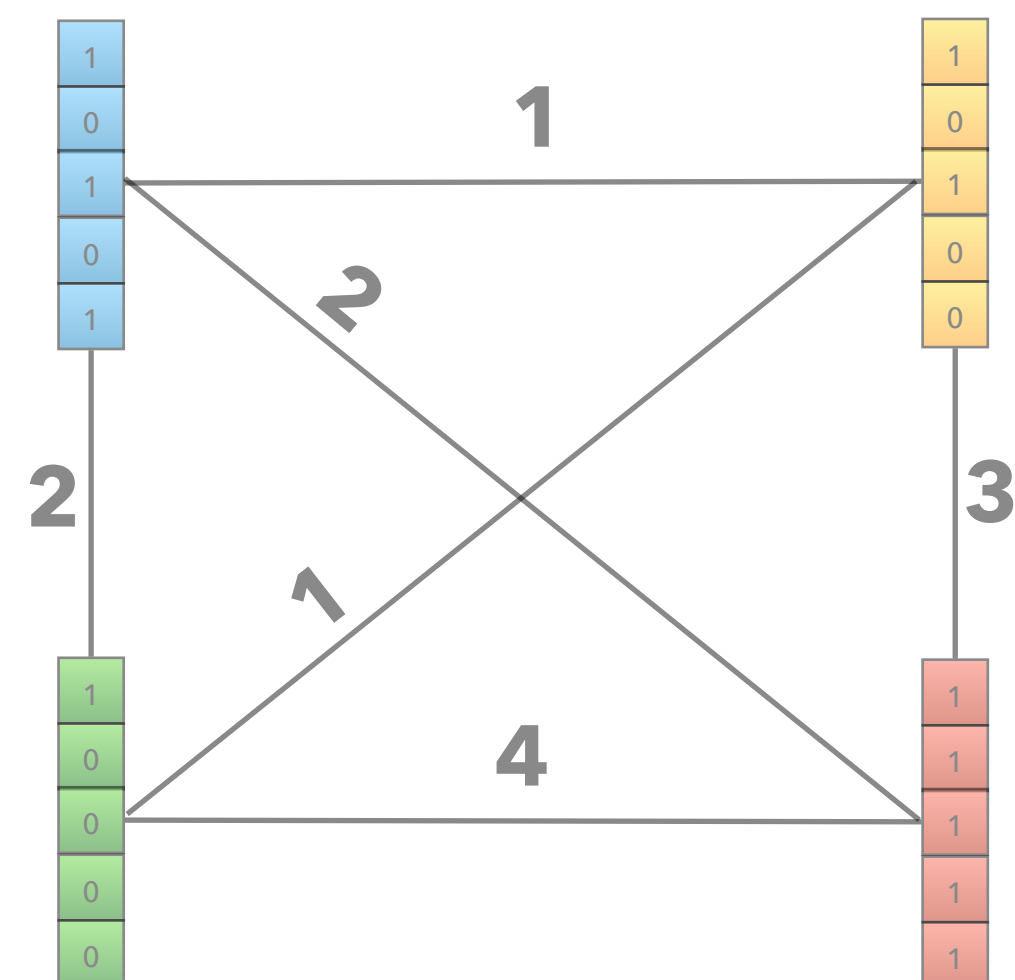
CCG derived from dbG



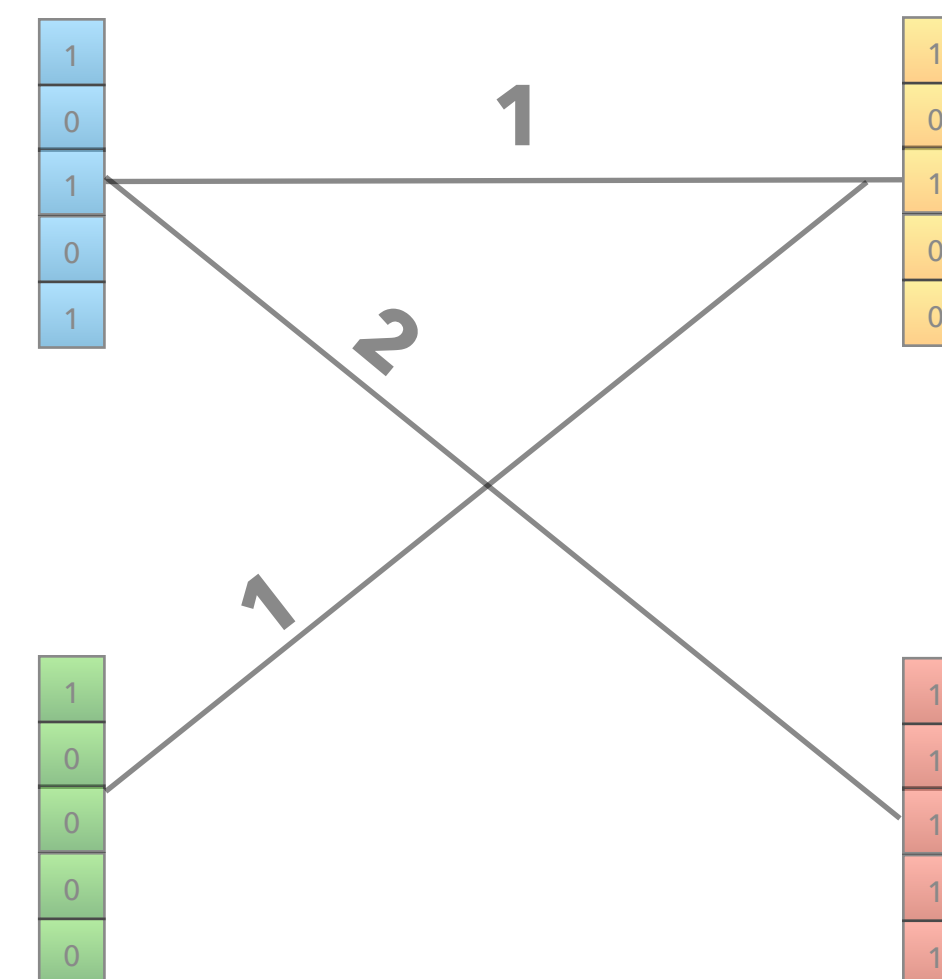
MST on our Graph



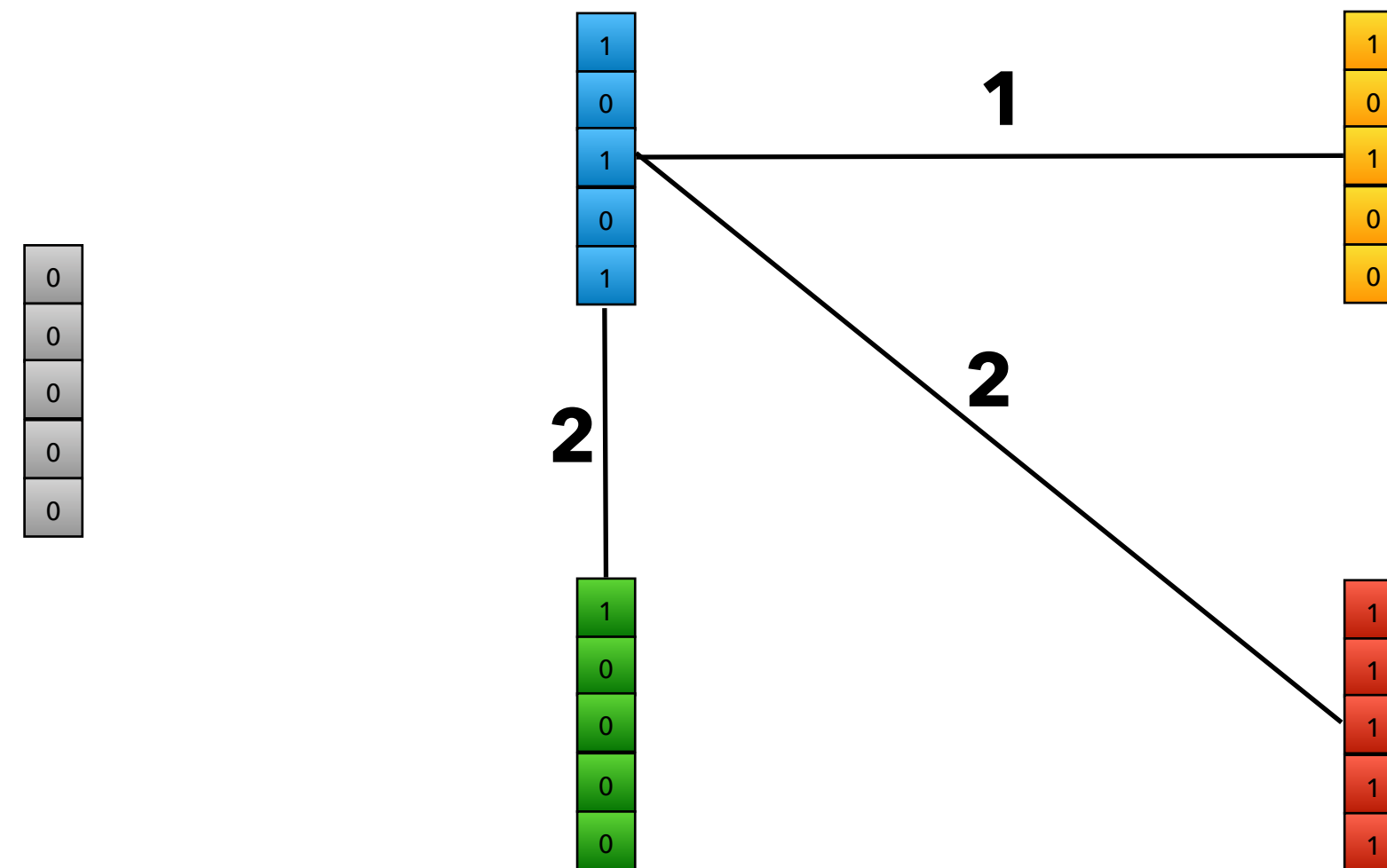
Complete CCG



Optimal MST

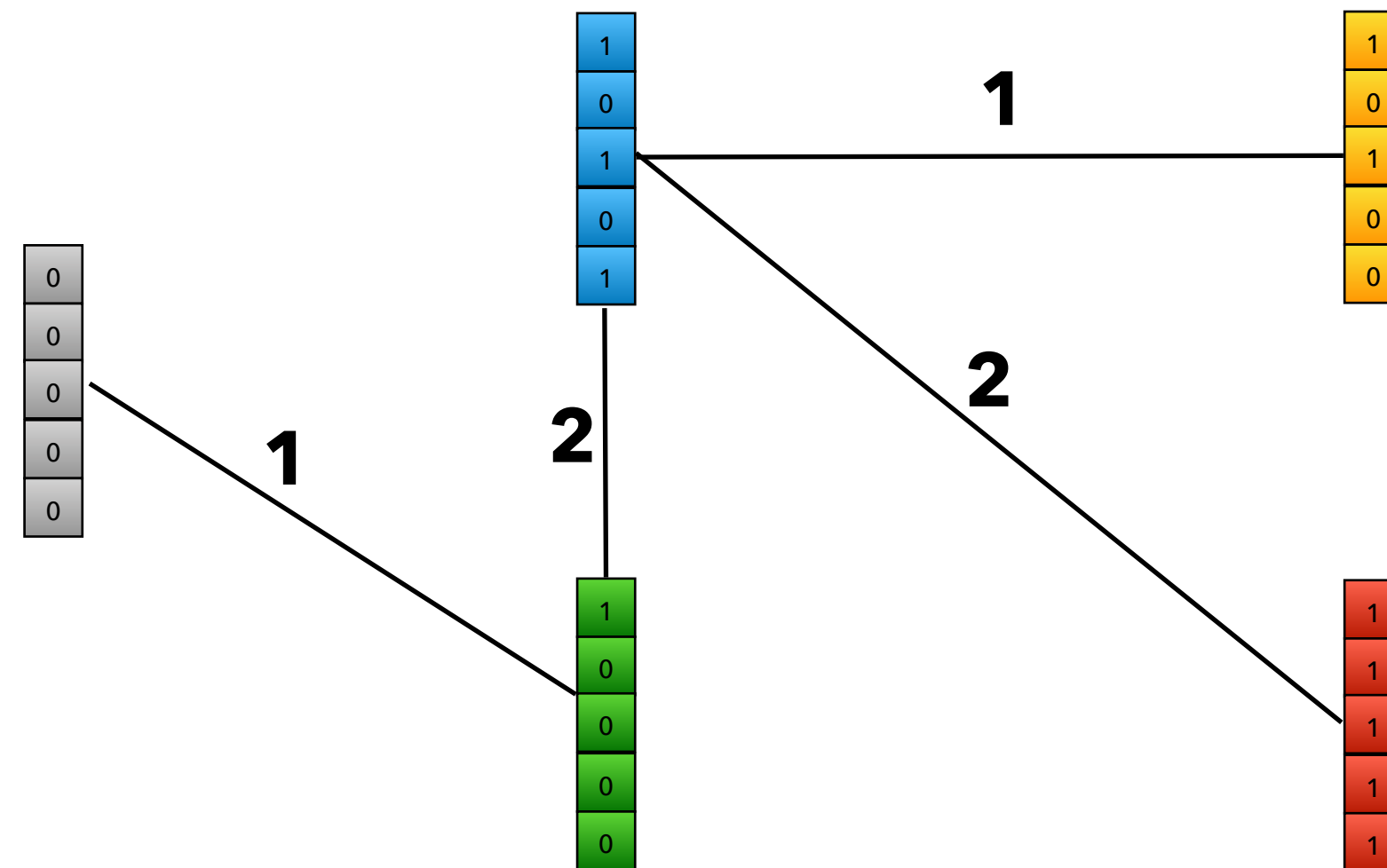


The MST efficiently encodes related color classes



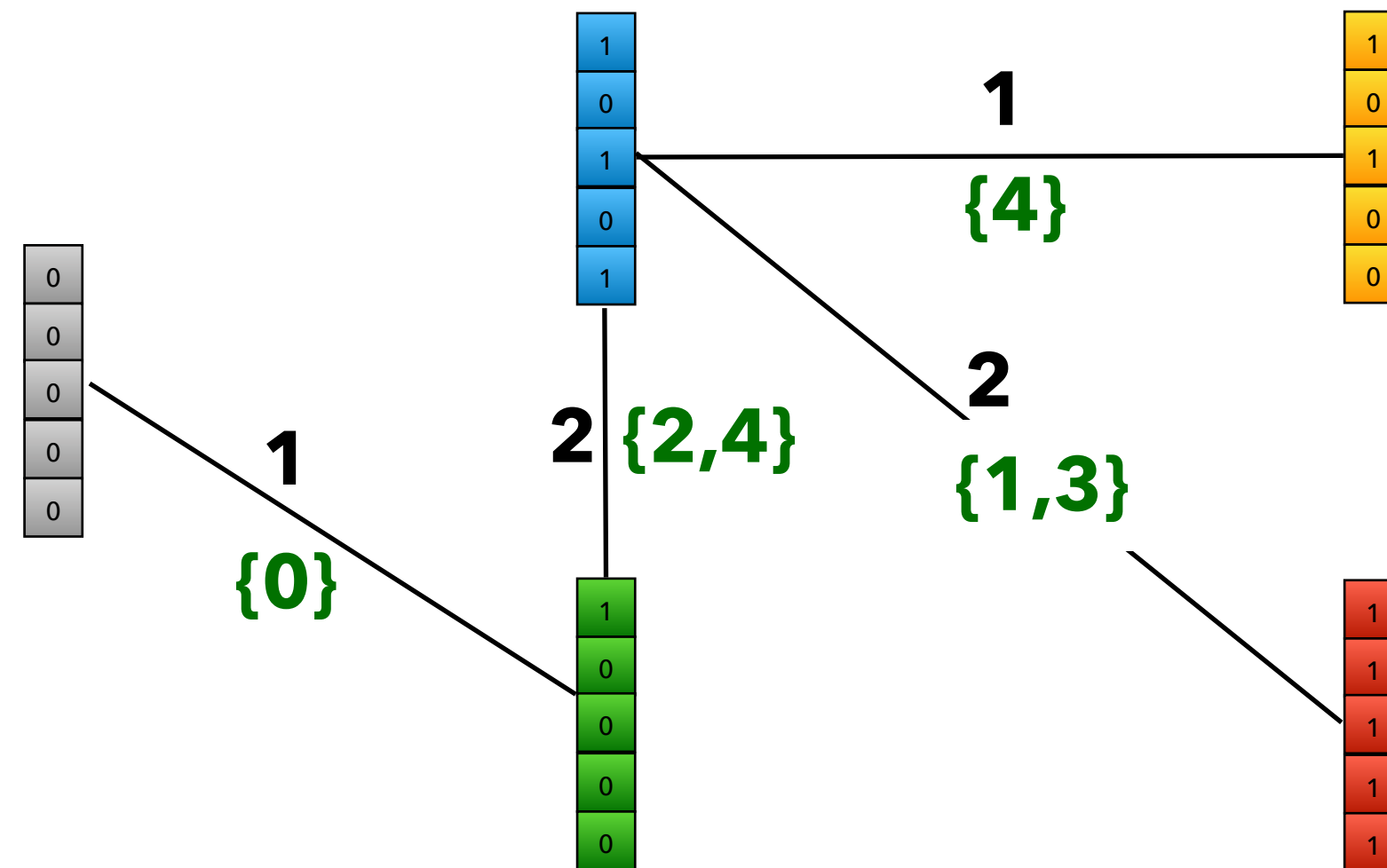
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Augment with all 0 color class to guarantee one, connected MST



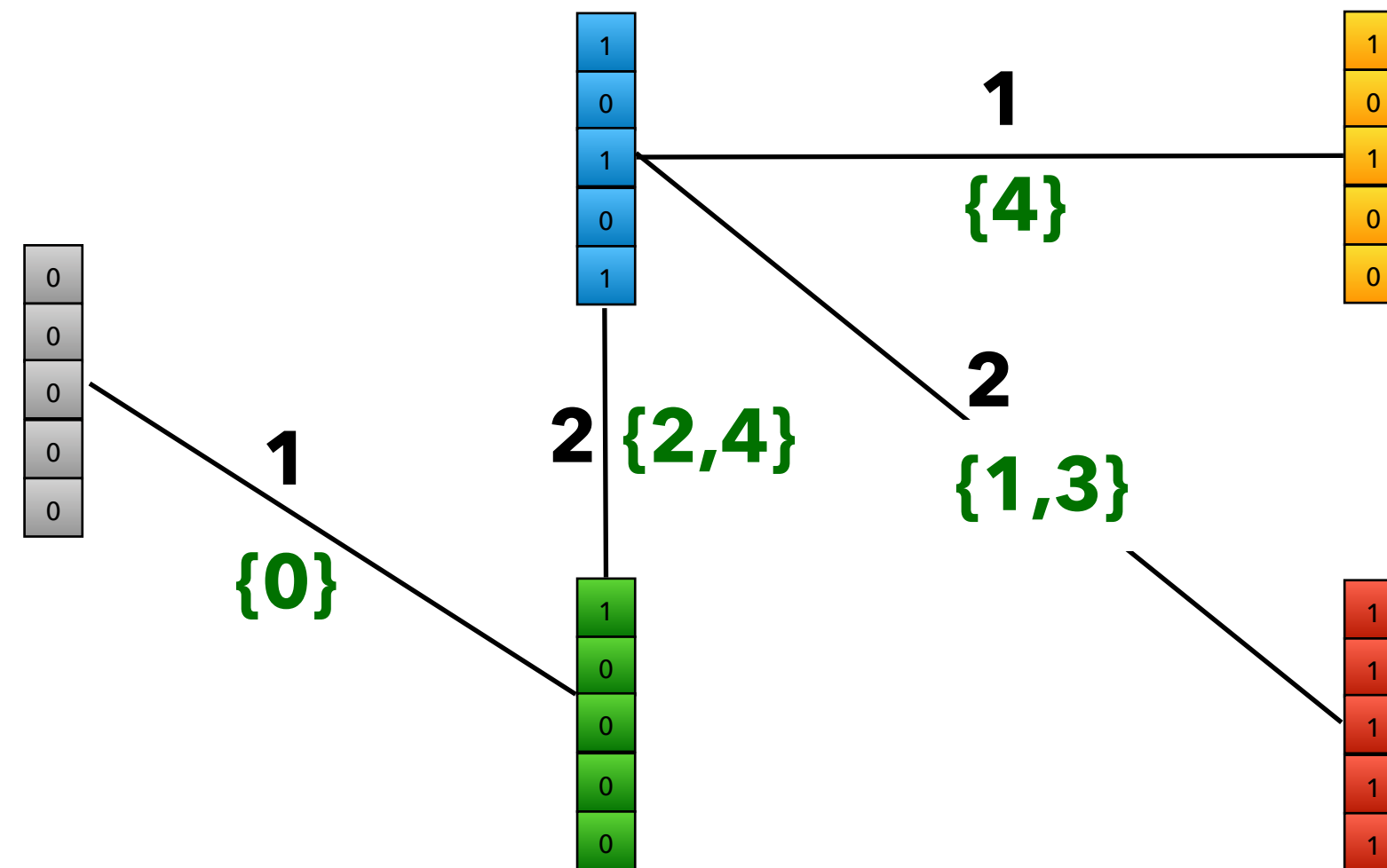
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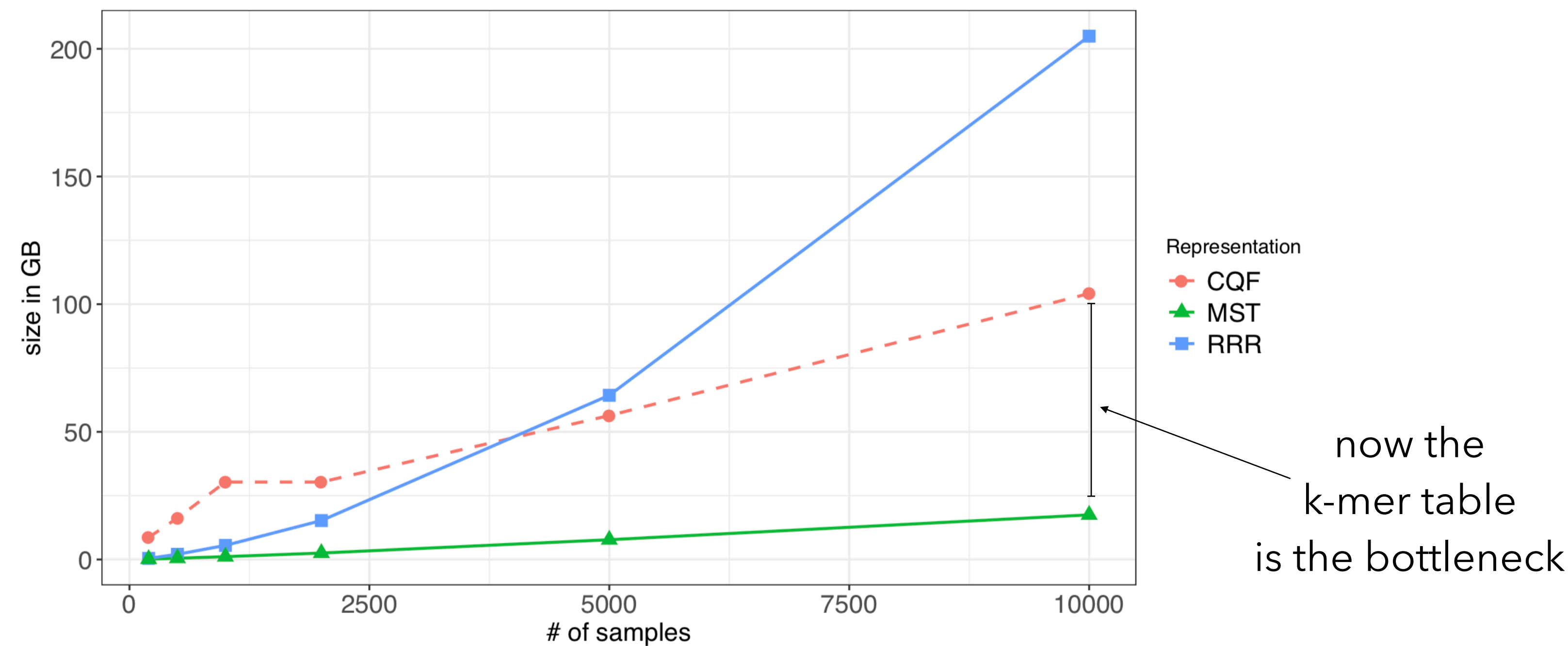
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To reconstruct a vector, walk from your node to the root, flipping the parity of the positions you encounter on each edge.

The MST approach scales very well



Dataset	# samples	MST					$\frac{\text{size}(MST)}{\text{size}(RRR)}$
		RRR matrix	Total space	Parent vector	Delta vector	Boundary bit-vector	
<i>H. sapiens</i> RNA-seq samples	200	0.42	0.15	0.08	0.06	0.01	0.37
	500	1.89	0.46	0.2	0.24	0.03	0.24
	1,000	5.14	1.03	0.37	0.6	0.06	0.2
	2,000	14.2	2.35	0.71	1.5	0.14	0.17
	5,000	59.89	7.21	1.72	5.1	0.39	0.12
	10,000	190.89	16.28	3.37	12.06	0.86	0.085
Blood, Brain, Breast (BBB)	2586	15.8	2.66	0.63	1.88	0.16	0.17

Improvement over RRR improves with # of samples

dataset from SBT / SSBT / Mantis paper

How does MST approach affect query time?

One concern is that replacing $O(1)$ lookup with MST-based decoding will make lookup slow; does it?

How does MST approach affect query time?

One concern is that replacing $O(1)$ lookup with MST-based decoding will make lookup slow; does it?

Turns out a caching strategy (an LRU over popular internal nodes) keeps it just as fast as lookup in the RRR matrix

	Mantis with MST			Mantis		
	index load + query	query	space	index load + query	query	space
10 Transcripts	1 min 10 sec	0.3 sec	118GB	32 min 59 sec	0.5 sec	290GB
100 Transcripts	1 min 17 sec	8 sec	119GB	34 min 33 sec	11 sec	290GB
1000 Transcripts	2 min 29 sec	79 sec	120GB	46 min 4 sec	80 sec	290GB

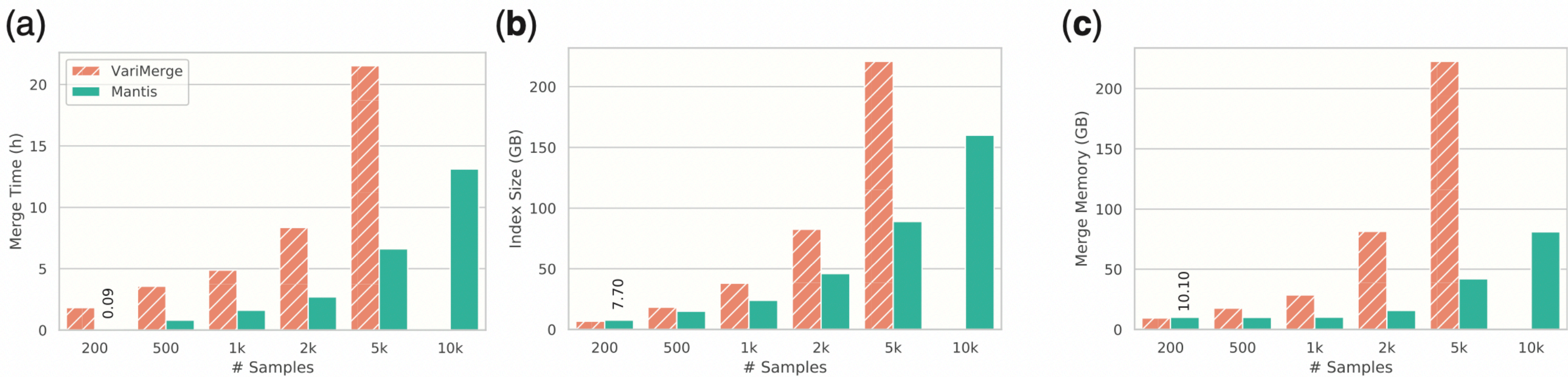
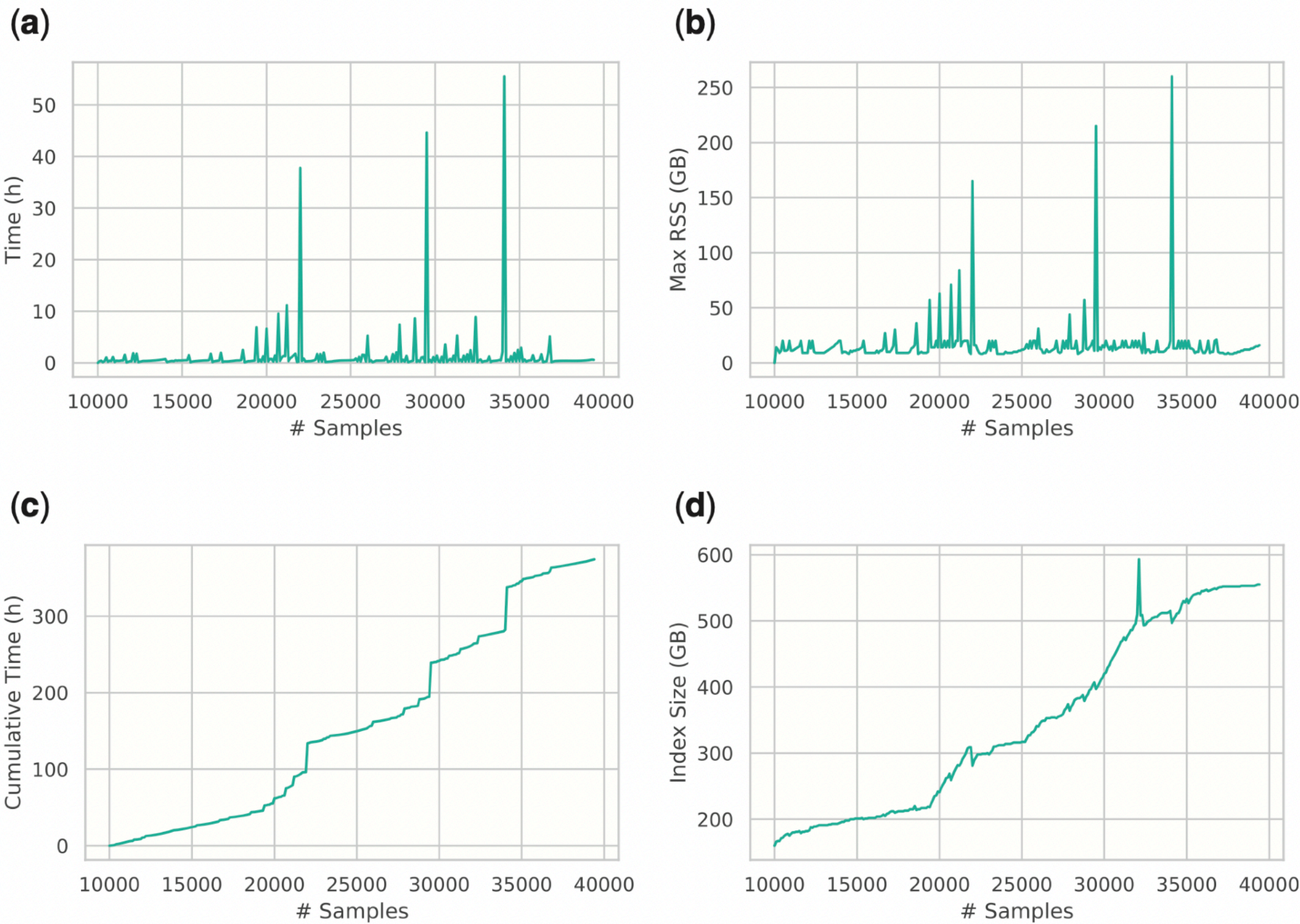
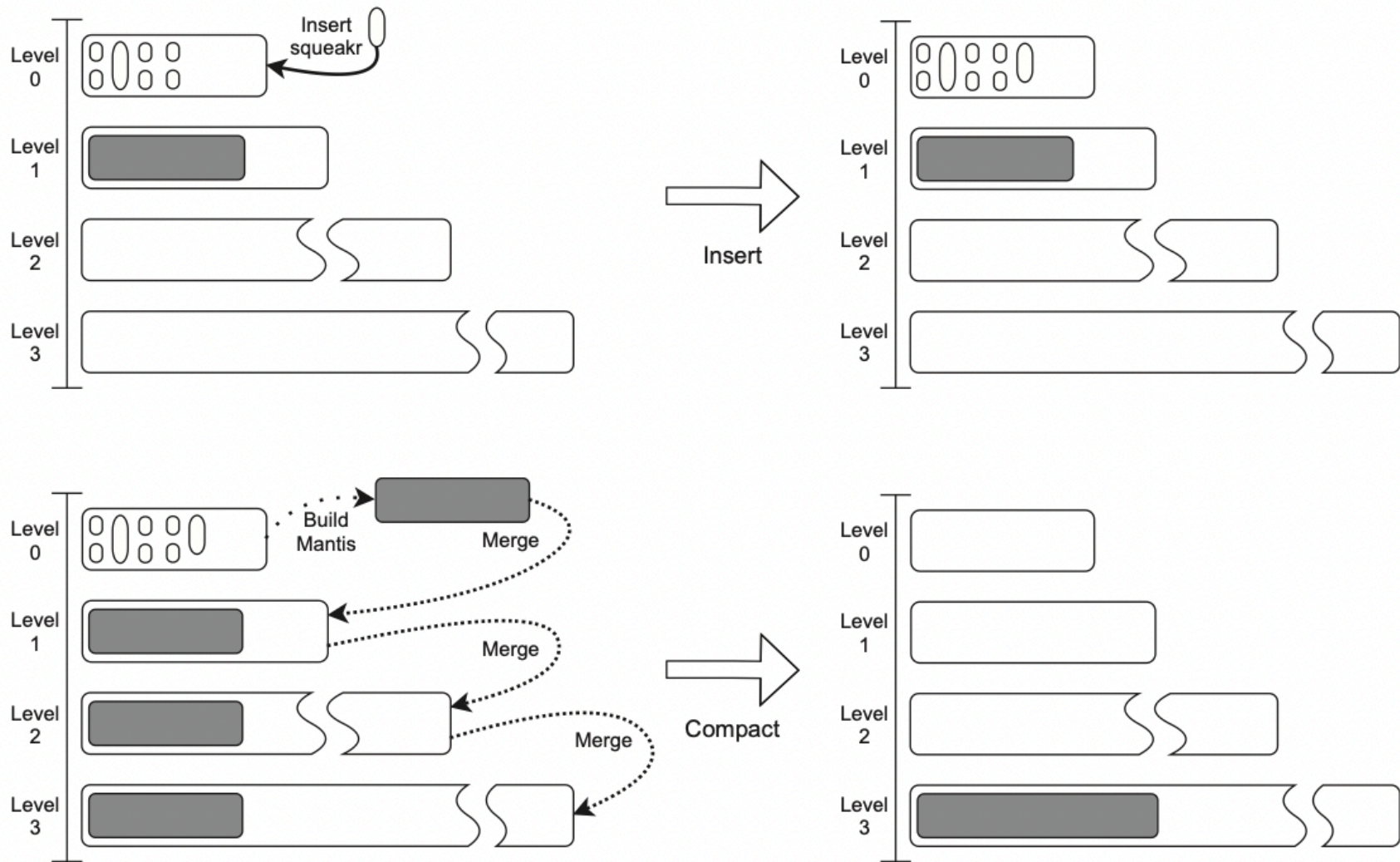
State-of-the-art marches on

Making the Mantis index dynamic

Sequence analysis

An incrementally updatable and scalable system for large-scale sequence search using the Bentley–Saxe transformation

Fatemeh Almodaresi ¹, Jamshed Khan ¹, Sergey Madaminov², Michael Ferdma Rob Johnson³, Prashant Pandey³ and Rob Patro ^{1,*}



Compressing generic annotations with the counting dBG

Lossless Indexing with Counting de Bruijn Graphs

Mikhail Karasikov^{1,2,3}, Harun Mustafa^{1,2,3}, Gunnar Rätsch^{1,2,3,4,5,*}, and André Kahles^{1,2,3,*}

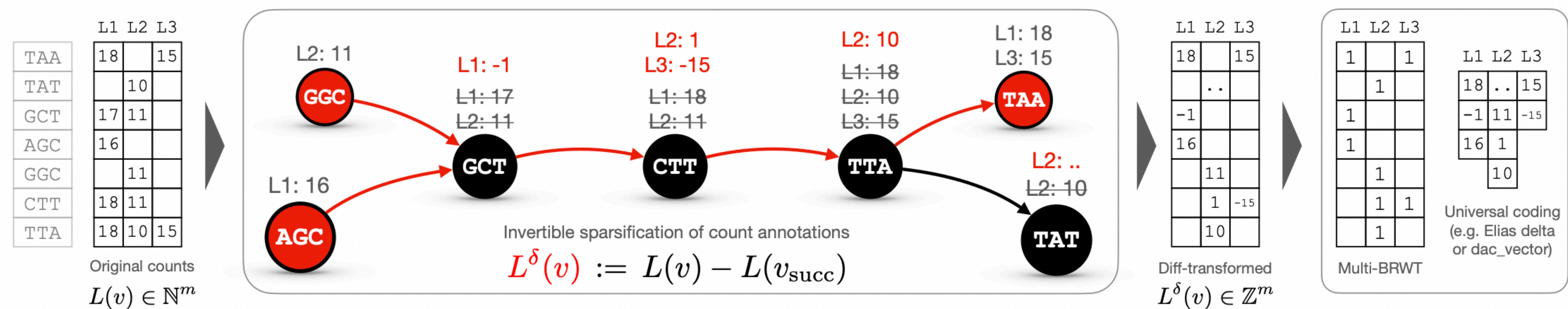


Table 3. Lossless indexing of RefSeq (rel. 97) with k-mer coordinates for the complete data set (32,881,422 accessions, 1.7 Tbp) and the set of all Fungi (69,034 accessions, 8.8 Gbp).

Method	RefSeq (Fungi)		RefSeq (All)	
MegaBLAST	12.3 GB	11.19 bits/bp	2,359 GB	11.07 bits/bp
This work	3.3 GB	2.97 bits/bp	533 GB	2.50 bits/bp

Note RefSeq here is still reference genomes, << smaller than all “raw data”, but notice the substantial space improvement over MegaBLAST. Also, the counting dBG allows actual alignment; not just k-mer matching

Metagraph

Article

Efficient and accurate search in petabase-scale sequence repositories

<https://doi.org/10.1038/s41586-025-09603-w>

Received: 7 June 2024

Accepted: 8 September 2025

Published online: 8 October 2025

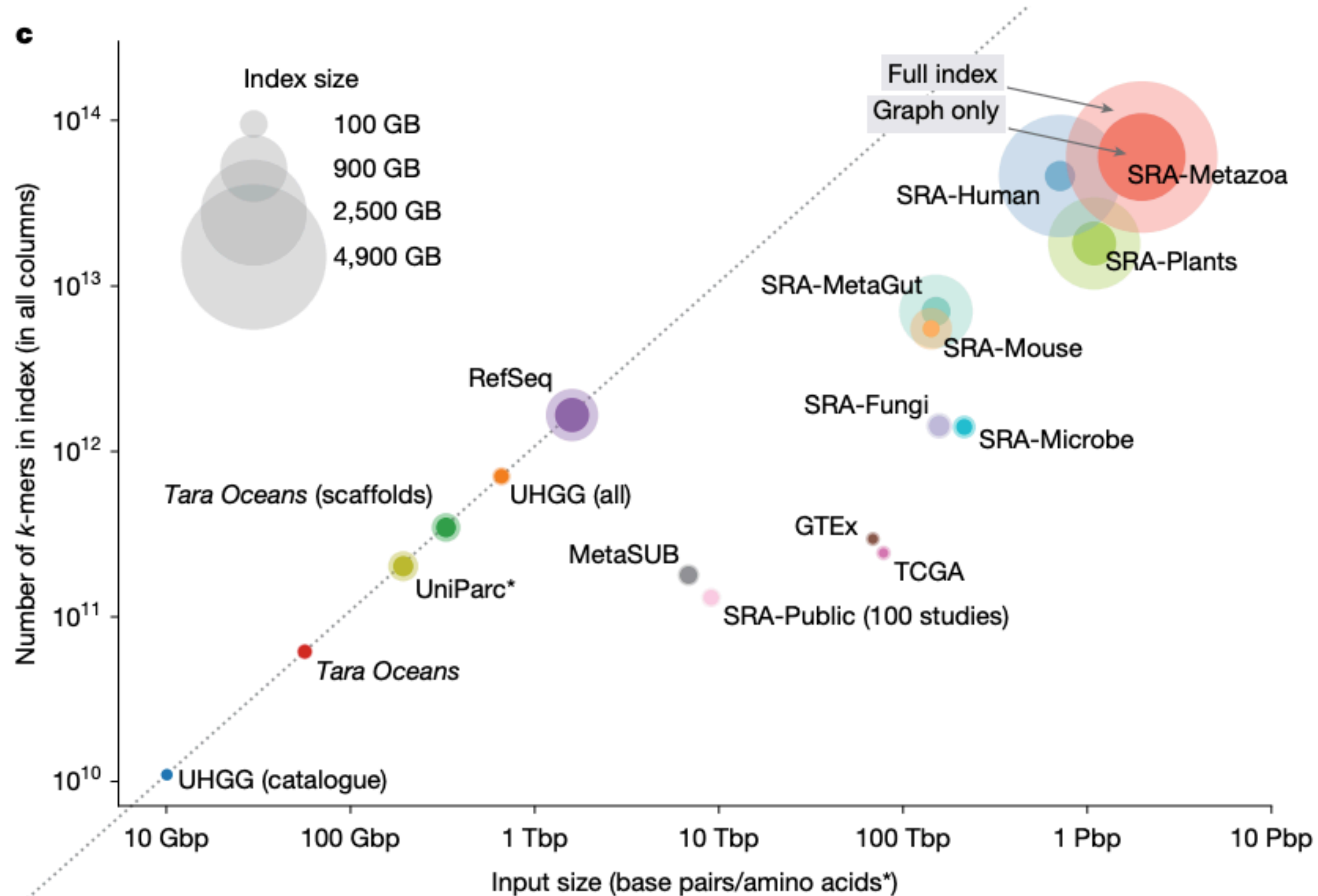
Open access

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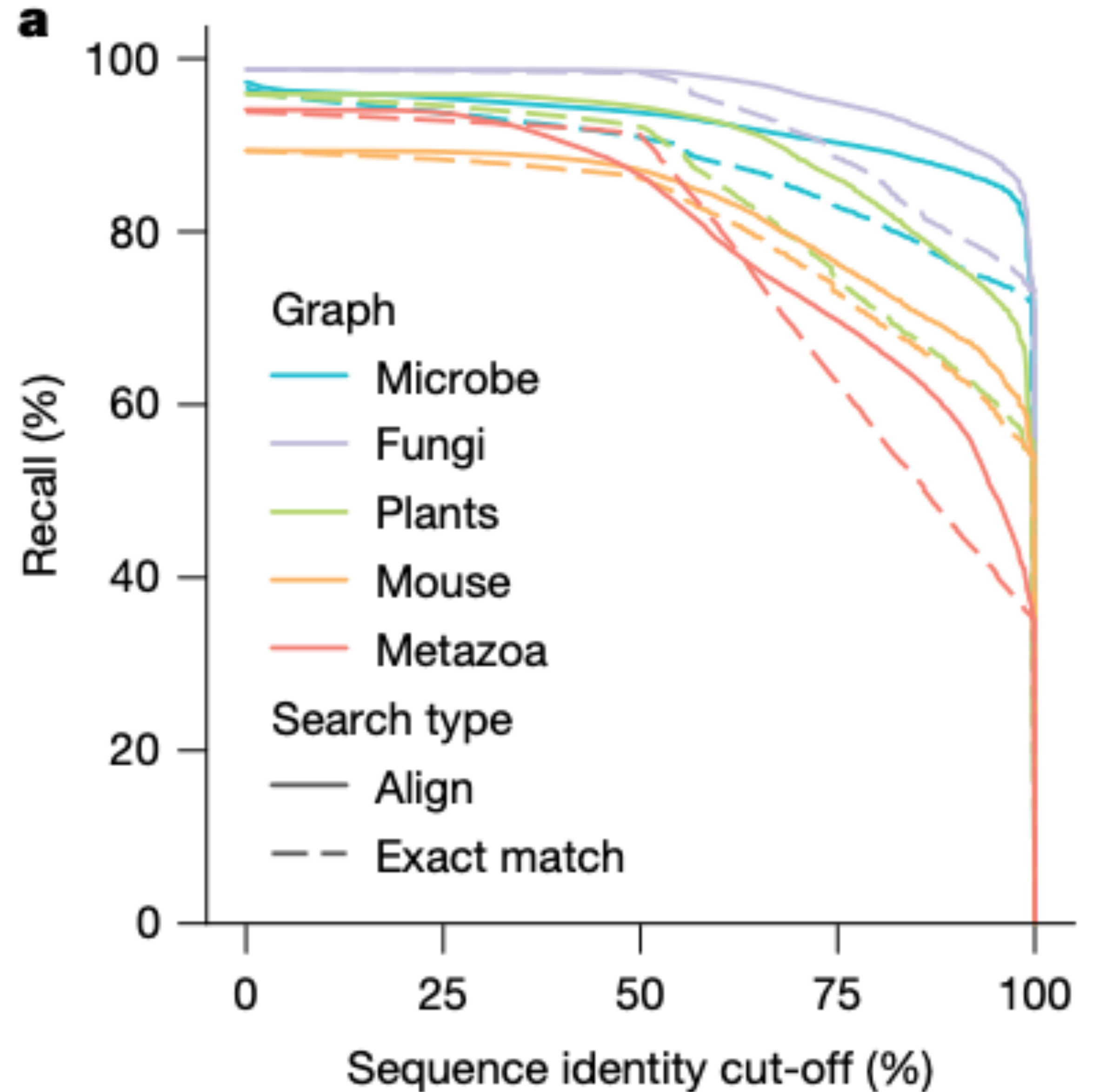
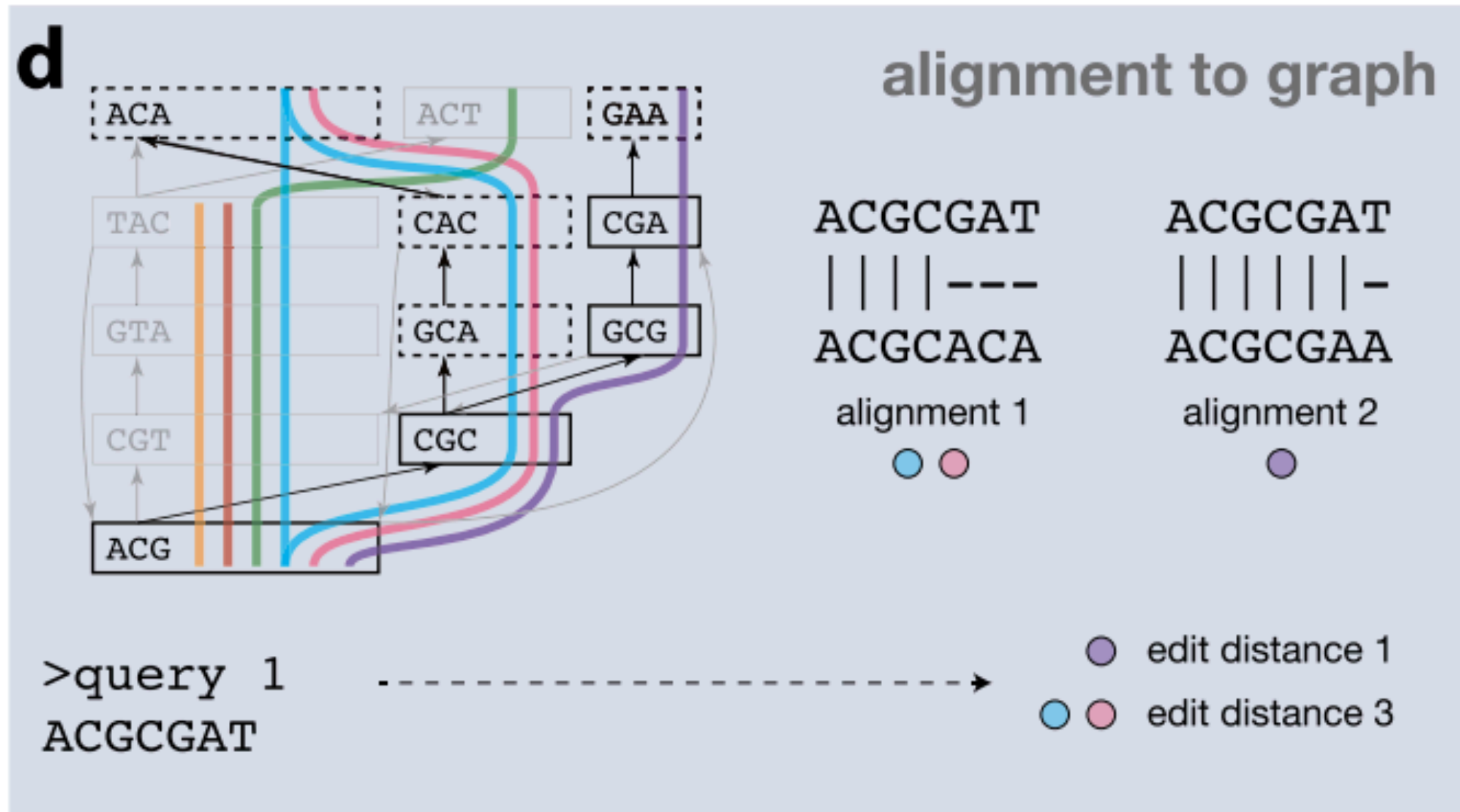
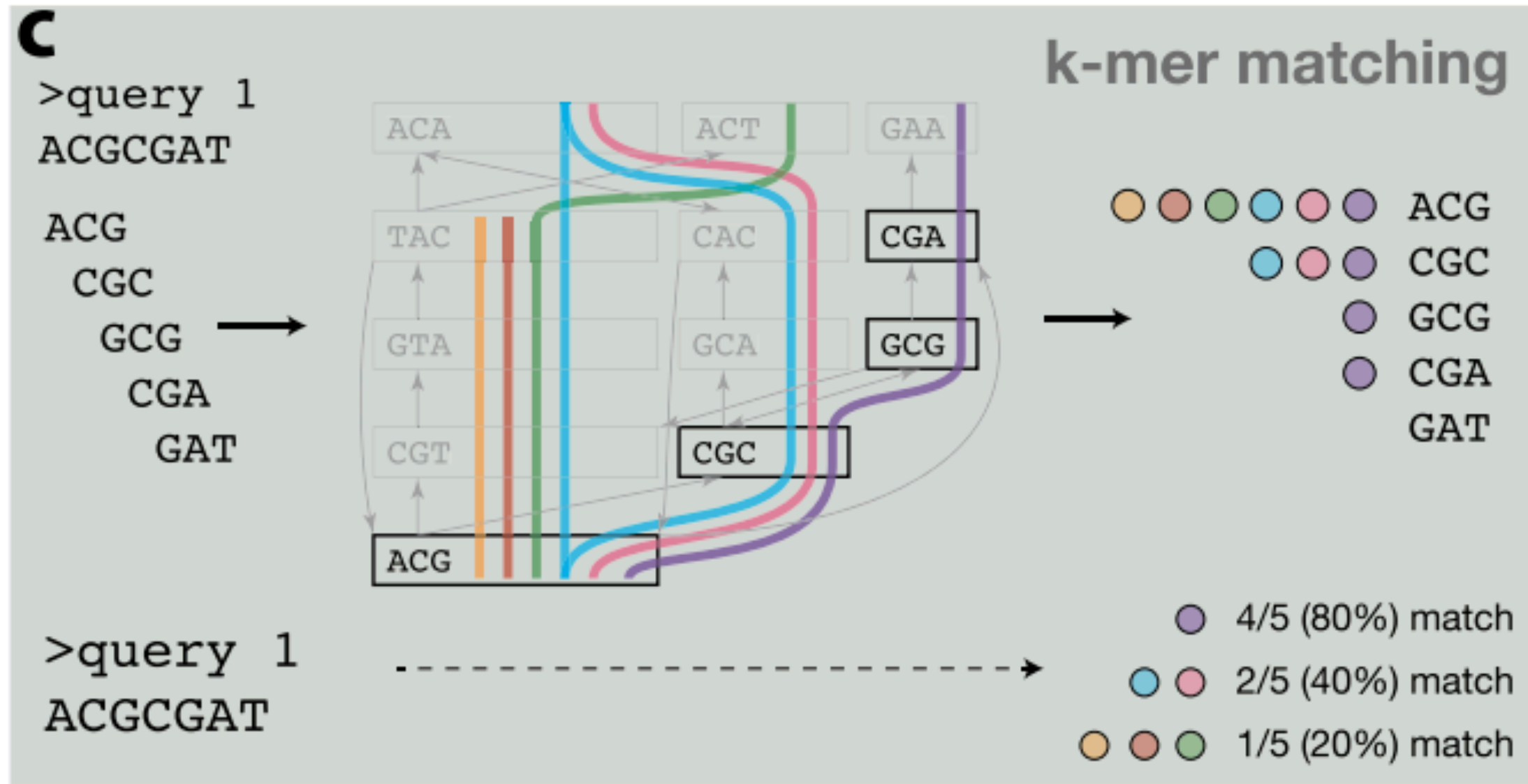
Mikhail Karasikov^{1,2,3,6}, Harun Mustafa^{1,2,3,6}, Daniel Danciu^{1,3}, Oleksandr Kulkov^{1,3}, Marc Zimmermann^{1,3}, Christopher Barber^{1,3}, Gunnar Rätsch^{1,2,3,4,5} & André Kahles^{1,2,3}✉

The amount of biological sequencing data available in public repositories is growing rapidly, forming a critical resource for biomedicine. However, making these data efficiently and accurately full-text searchable remains challenging. Here we build on efficient data structures and algorithms for representing large sequence sets^{1–6}. We present MetaGraph, a methodological framework that enables us to scalably index large sets of DNA, RNA or protein sequences using annotated de Bruijn graphs. Integrating data from seven public sources^{7–13}, we make 18.8 million unique DNA and RNA sequence sets and 210 billion amino acid residues across all clades of life—including viruses, bacteria, fungi, plants, animals and humans—full-text searchable. We demonstrate the feasibility of a cost-effective full-text search in large sequence repositories (67 petabase pairs (Pbp) of raw sequence) at an on-demand cost of around US\$100 for small queries up to 1 megabase pairs (Mbp) and down to US\$0.74 per queried Mbp for large queries. We show that the highly compressed representation of all public biological sequences could fit on a few consumer hard drives (total cost of around US\$2,500), making it cost-effective to use and readily transportable for further analysis. We explore several practical use cases to mine existing archives for interesting associations, demonstrating the use of our indexes for integrative analyses, and illustrating that such capabilities are poised to catalyse advancements in biomedical research.

Metagraph: A collection of indices



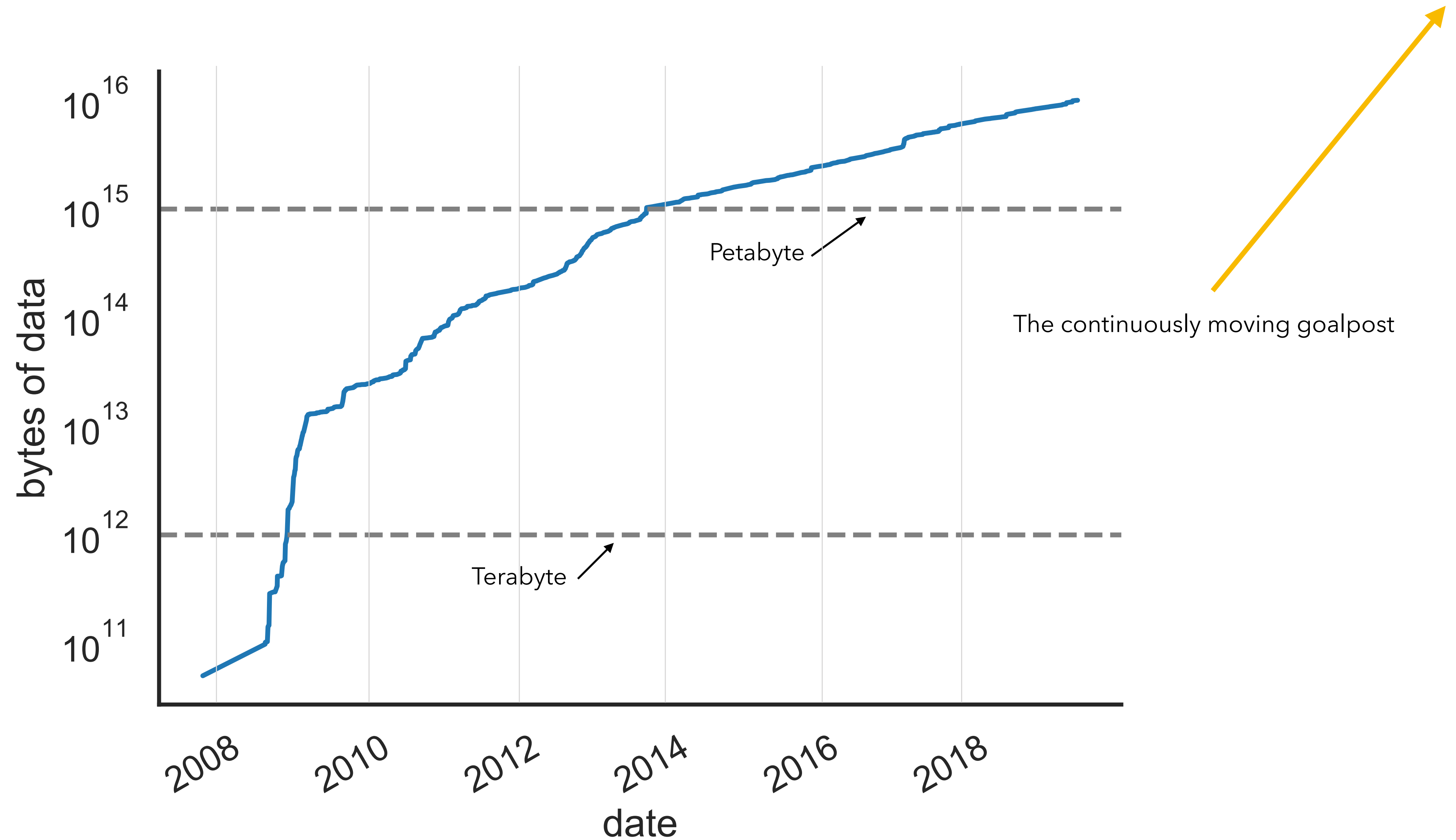
Metagraph: A collection of indices



Metagraph: Some remaining challenges

- Indices are partitioned, but not independently “updatable”, to add new experiments, you must build a new index over them
- Query in RAM is fast (<1s) , but if the index doesn't fit in RAM, point queries are *slow* (10s of min)
- Resources for *constructing* the index are vastly larger than those to serve the index once built — metagraph has a build then compress paradigm — this leads to a necessarily heterogeneous system where, to avoid wasting resources, the nodes that build indices must not be the nodes that serve them.

A Call To Arms



"It seems that some essentially new ... ideas are here needed"
– Paul Adrien Maurice Dirac*